Virtuoso® Layout Pro: T8 Virtuoso Concurrent Layout Editing

Course Version IC 25.1

Lecture Manual **Revision 1.0**

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Virtuoso Layout Pro: T8 Virtuoso Concurrent Layout Editing

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Module 1

About This Course

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Course Prerequisites

Before taking this course, you need to have already completed the following courses:

- Virtuoso® Layout Pro: T1 Environment and Basic Commands
- Virtuoso Layout Pro: T2 Create and Edit Commands
- Virtuoso Layout Pro: T3 Basic Commands
- Virtuoso Layout Pro: T4 Advanced Commands
- Virtuoso Layout Pro: T5 Interactive Routing
- Virtuoso Layout Pro: T6 Constraint-Driven Flow and Power Routing
- Virtuoso Layout Pro: T7 Module Generator and Floorplanner

Or you need to have:

- Experience with layout design
- Knowledge of schematic symbols and MOS devices
- A basic knowledge of UNIX/Linux

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Course Objectives

In this course, you

- Set up and analyze the Concurrent Layout Editing environment
- Initialize and partition the top design in Manager mode
- Edit the design partitions in designer mode
- Merge and commit the top design in Manager mode

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Course Agenda

- Setting Up the Concurrent Layout Editing Environment
  - Setting Up the Working Directory
- Initializing and Partitioning the Top Design in Manager Mode
  - Initializing the Top Design in Manager Mode
  - Partitioning the Top Design in Manager Mode
- Editing the Design Partition userA in Designer Mode
  - Setting the Edit Scope for the Design Partition userA in Designer Mode
  - Editing the Design Partition userA Using the CLE Flow
  - Editing the Design Partition userA Hierarchically Using the CLE Flow
- Editing the Design Partition userB in Designer Mode
  - Setting the Edit Scope for the Design Partition userB in Designer Mode
  - Editing the Design Partition userB Using the CLE Flow
  - Editing the Design Partition userB Hierarchically Using the CLE Flow
- Merging/Committing the Top Design in Manager Mode
  - Merging/Committing the Top Design in Manager Mode

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Software and Licenses

For the software and licenses used in the labs for this course, go to:

https://www.cadence.com/en_US/home/training/all-courses/86302.html

If there is additional information regarding the specific software, it is detailed in the lab document and/or the README file of the database provided with this course.

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- Log in to our Learning Management System, click on the course in your transcript, and go to the **Content** tab to locate the exam.

How long will it take to complete the exam?

- Most exams take 45 to 90 minutes to complete. You may retake the exam multiple times to pass the exam.

How do I access and use the digital badge?

- After you pass the exam, you get a digital badge and instructions on how to place it on social media sites.

How is the digital badge validated?

- Credly validates the digital badge as issued to you by Cadence and includes the details of the criteria you completed to earn the badge.

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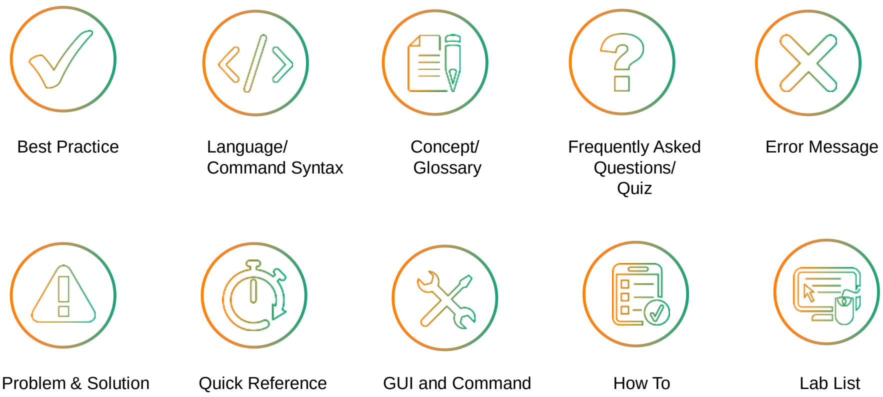
Icons Used in This Class



Throughout this class, we use icons to draw your attention to certain kinds of information. Here are the icons we use, and what they mean.

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Module 2

Setting Up the Concurrent Layout Editing Environment

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Module Objectives

In this module, you

- Set up the Concurrent Layout Editing environment
- Analyze the default Virtuoso® Layout flow and the Concurrent Layout flow
- List the Concurrent Layout modes
- Examine the Hierarchical Edit modes in CLE

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Terms Used in Concurrent Layout Editing

Area-based design partition	Defines one or more areas for a designer to edit within.
Design manager	Defines the design partition for each designer and merges the respective design partition views back to the top design.
Design partition	Divides the design responsibilities among designers. You can create as many partitions as needed in a design, and new design partitions can be created at anytime.
Design partition view	The on-disk layout cellview that stores incremental edits to the top design separately. These updates can be reapplied to the same top design later.

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Terms Used in Concurrent Layout Editing (continued)

Designer	A user who edits in the assigned design partition.
Free design partition	Allows the user to work on any object, net, and any part of the design.
Net-based design partition	Defines a net set for a designer to edit only those objects that have allowed or no connectivity.
Object-based design partition	Defines the object ownership for a designer to edit. The person who created the object also owns it.

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Terms Used in Concurrent Layout Editing (continued)

Peer designer	A designer working on another design partition of the same top design concurrently. Updates made by the peer designer are saved to a different design partition view.
Top cellview (or top design)	This is the initial layout cellview to be concurrently edited after initialization. After opening in memory, it is called the top design.

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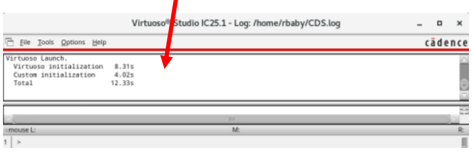
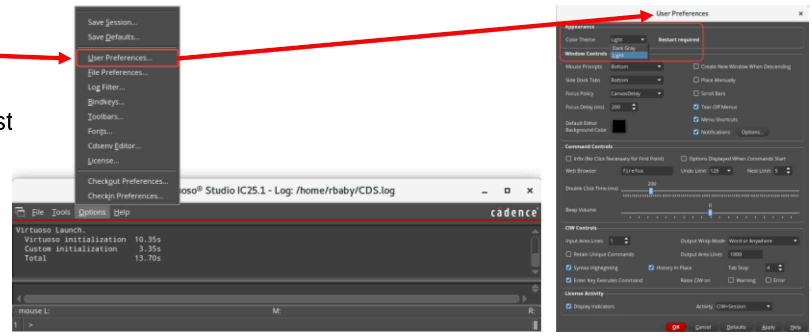
Switching the Color Theme of the Virtuoso Studio Session

From the CIW, select **Options – User Preferences** and use the *Appearance* section at the top of the form to choose the color theme for the Virtuoso Studio session.

- In IC25.1, you can select between two themes:
  - Dark Gray (default)
  - Light
- The theme change is applied after you rest the session, as shown here.





The appearance setting is user-specific and stored in *~/.cadence/ui/theme.json* file.

Virtuoso Studio Themes

Virtuoso Studio provides the flexibility to change color themes. For some, visual performance tends to be better when viewing the design in a Light theme and for some using the Dark Gray theme generates a better visual experience.

Switching to the Dark Gray theme reduces power usage, improves visibility, and makes it easier for designs to be read. In this theme, the background is Dark Gray, and the text uses a contrasting lighter color. The Dark Gray theme also introduces modern visual effects such as mouse hover, smaller sidebar and flat design.

By default, the CIW opens in dark mode. To switch between color themes, use the Color Theme option in the Appearance section of the User Preferences form.

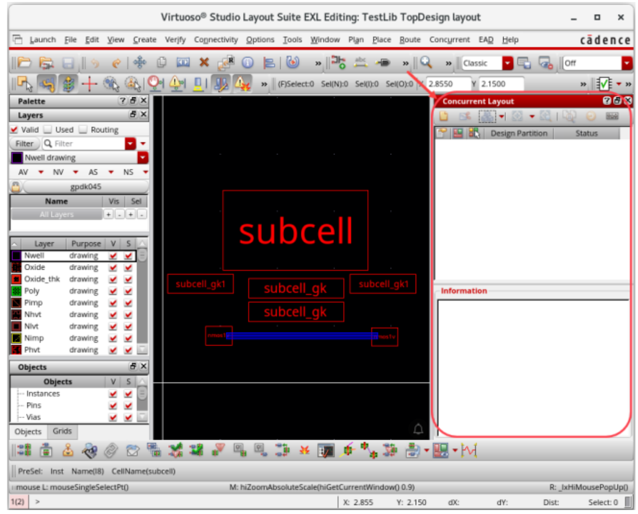
The Dark Gray theme is introduced for VSE, VLS and CIW, and for all Cadence-style windows such as Library Manager, SKILL API Finder, ADE Maestro, and Pegasus.

What Is the Concurrent Layout Editing?

The Virtuoso Concurrent Layout (CLE *with E for editing*) is a layout editing environment that enables designers to work concurrently on the same cellview within Virtuoso.

As the designers work in parallel, using the Concurrent Layout helps in reducing the designing time and, as a result, increases productivity.

- You can perform concurrent editing in Layout XL, Layout EXL, and Layout MXL.



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Licensing Requirements to Access the Concurrent Layout Functionality

The Concurrent Layout functionality requires the following licenses:

- A `Virtuoso_Layout_Suite_XL` license combined with 4 GXL flexible license tokens
- A `Virtuoso_Layout_Suite_EXL` license
- A `Virtuoso_Layout_Suite_MXL` license
  - The license is held until the last layout window is closed.
    - If you switch from Layout XL to Layout EXL or to Layout MXL, the GXL tokens remain checked out.
      - To prevent this, first close Layout XL, and then open Layout EXL or Layout MXL.

The Concurrent Layout functionality is available from **ICADVM18.1** onwards.

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Benefits of Using the Concurrent Layout

Some of the top benefits of using the Concurrent Layout include the following:

- Boosts layout productivity by enabling several designers to work concurrently on the same cellview.
  - Typical examples are DRC fixing, chip finishing, and critical nets manual routing.
- Saves the design partition view containing only the updated part of the design, which is quite small in comparison to the initial cellview.
  - This reduces the disk access time.
- Enables the design manager to review a partitioned view and merge or reject it.
  - The user can generate several results for what-if analysis and picks the best combination in the end.
- Supports offline concurrent editing because the client/server model can suffer synchronization bottleneck over the network.
- Provides incremental Edit In Place to complement the traditional hierarchical design by postponing an update in the sub-hierarchy until it is verified in all the designs referencing it.

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Limitations of Using the Concurrent Layout

Some tasks that are currently not supported in the Concurrent Layout include the following:

- Constraint editing:
  - For example, you cannot edit MODGENs in the Concurrent Layout.
- Editing an object created in an imported peer partition:
  - Edits made to such objects are not saved. You can only concurrently edit an object existing in the top design.

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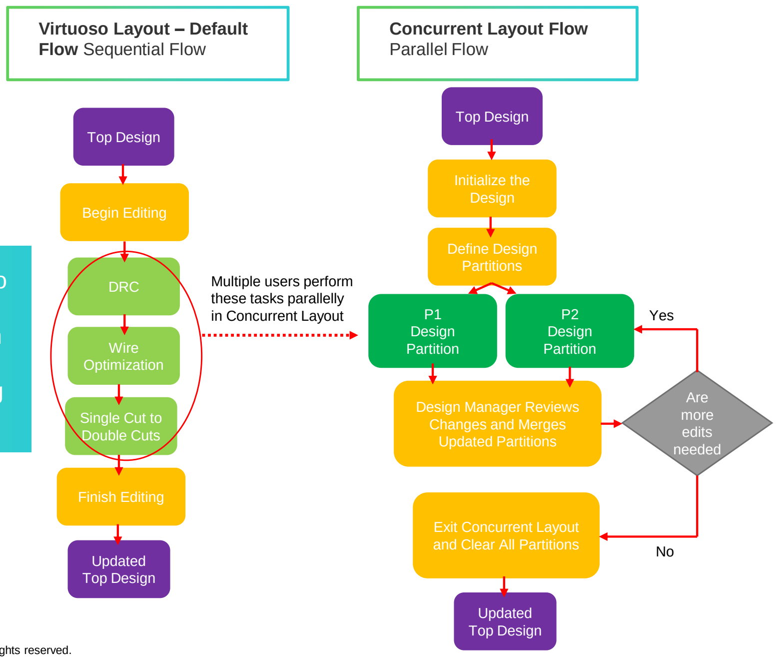
The Default Virtuoso Layout Flow and Concurrent Layout Flow



The default Virtuoso Layout flow is sequential, in which each task is carried out after completing the previous task.

In the CLE environment, you can create multiple partitions in the design and have designers work in parallel, which helps save time.

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The Default Virtuoso Layout Flow Tasks

The default Virtuoso Layout flow involves the following tasks in sequence:

1. Open the top design.
2. Begin the editing.
3. Perform various tasks such as DRC checks, wire optimization, making a single cut to double cuts, and so on.
4. Finish the editing.
5. Open the updated top cellview in the Virtuoso layout.

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The Concurrent Layout Flow Tasks

The basic Concurrent Layout flow involves the following tasks in parallel:

1. Open the top design. (Design Manager)
2. Initialize the design for Concurrent Layout Editing. (Design Manager)
3. Define design partitions based on how you want to divide the work among various designers. (Design Manager)
4. Perform various tasks such as DRC checks and wire optimization in parallel on assigned designed partitions. (Designers)
5. Save design partition updates in respective design partition views and submit the updates for merge with the top design. (Designers)
6. Review and merge design partitions. (Design Manager)
7. Repeat steps four to six as needed. A merged design partition is auto reset in memory when it is reopened. (Designers)
8. Exit the CLE environment and clear all design partitions. (Design Manager)
9. Open the updated top cellview in the Virtuoso layout. (Design Manager)

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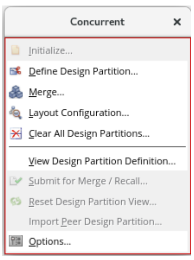
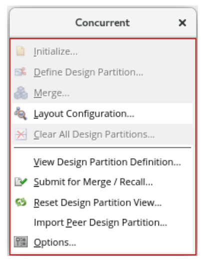
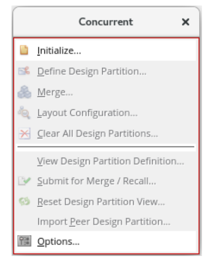
Concurrent Layout Modes

The commands that are enabled in the Concurrent menu depend on the mode in which the design is open. These modes are the *Manager mode*, the *Designer mode*, and the *Single user mode*.

Manager Mode	Designer Mode	Single User Mode
Lets you perform various managerial tasks, such as defining the design partition for each designer and merging the respective design partition views back to the top design.	Lets you edit the design in the assigned design partition view and then submit these changes for merging with the top design.	Lets a single user edit a design in a safe environment. You can choose to work on the complete design or in a focused area or focused range of layers. Your changes are implemented only when you save your changes and merge them with the top design.







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What Is Hierarchical Editing in the CLE Flow?

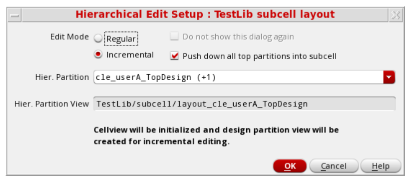
Hierarchical editing in the CLE flow lets multiple designers to concurrently Edit In Place (EIP) into hierarchical subcells.

The edit scope is retained in both the area-based and layer-based design partitions.

- In area-based design partitions, area boundaries defined at the top design partition are pushed down the hierarchy.
- In layer-based design partitions, the edit scope is limited to the layers available in the top design partition.

Hierarchical Edit Modes in CLE:

- When you EIP into a hierarchical subcell from a top design partition view, CLE provides two hierarchical editing modes.
  - Regular
  - Incremental



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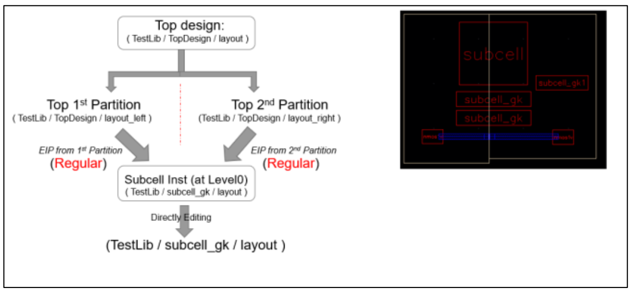
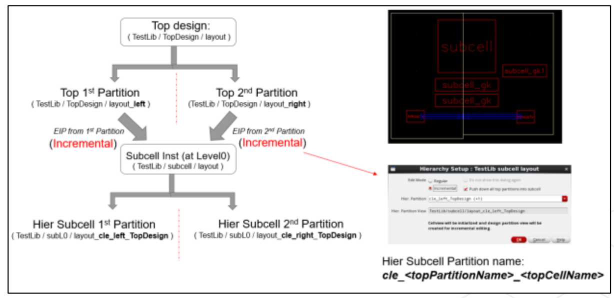
Hierarchical Edit Modes in CLE

Regular Edit Mode	Incremental Edit Mode
EIP in this mode lets you edit the main cellview, and only one user can make the edits at a time.	EIP in this mode keeps the original hierarchical subcell unmodified.
This means that concurrent editing of the hierarchical subcell is not possible in this mode.	A hierarchical design partition view is created when you EIP from the parent top design partition view.
After the subcell is edited and saved, the changes cannot be reversed.	The area partitions are pushed down in the hierarchical subcell from the parent top design partition view to keep the same edit scope in the hierarchical subcell.





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Roles and Responsibilities of the User in the CLE Flow

A user can play the following two roles in the CLE flow, namely:

- Manager
- Designer

The responsibilities they follow are based on their current role.

The **Manager** is the owner of the cellview. The manager has the following responsibilities:

- Decide whether to enable concurrent layout editing on the cellview.
- Distribute the cellview by defining design partitions based on the task.
- Review the progress of the developments in the respective design partitions without blocking them. This is called the health-check merge flow.
- Review and merge the edited design partitions submitted by the designers with the top design.

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Roles and Responsibilities of the User in the CLE Flow (continued)

A **Designer** is a user or users who work concurrently in the assigned design partition views. The designer has the following responsibilities:

- Edit scope for the designers is defined by the design partition region assigned to each design partition view.
  - Alerts are displayed if a designer edits outside the current design partition or when a peer designer tries to edit a design partition assigned to the current designer.
  - At any point during editing, designers can preview or import the changes made by peer designers to view updates made by them.
- While editing in a design partition view, a designer can either save the changes locally or check the updated design partition view.
  - The status of the design partition view stays as *Not Submitted*.
  - Manager can merge these changes to preview them but cannot permanently save them in the top design.
- After completing the edits, the designer needs to submit the edited design partition view for merge.
  - This changes the status of the design partition view to *Submitted*.
  - The manager can now review and merge the changes and then permanently save them in the top design.

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Module Summary

In this module, you learned how to

- Set up the Concurrent Layout Editing environment
- Analyze the default Virtuoso Layout flow and the Concurrent Layout flow
- List the Concurrent Layout modes
- Examine the Hierarchical Edit modes in CLE

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Lab

Lab 2-1 Setting Up the Working Directory

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Quiz: VLS Tier Support for Concurrent Editing

You can perform concurrent editing in VLS-L/XL/EXL/MXL tiers of Virtuoso software.

A. True

B. False

Answer: B

No. You can perform concurrent editing, in Layout XL, Layout EXL, and Layout MXL.

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Quiz: Virtuoso Release Support for the Concurrent Layout Functionality

In which of the following Virtuoso releases is the Concurrent Layout functionality available? Select all that apply.

A. ICADVM18.1
B. ICADVM20.1
C. IC23.1
D. IC25.1
E. IC6.1.8
F. All of the above

Answer: A, B, C, and D

The Concurrent Layout functionality is available in the Virtuoso ICADVM18.1, ICADVM20.1, IC23.1, and IC25.1 releases.

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Quiz: The Default Virtuoso Layout Flow and Concurrent Layout Flow

The default Virtuoso Layout flow is sequential, and each task is carried out after completing the previous task.

In the CLE environment, you can create multiple partitions in the design and have designers work in parallel, which helps save time.

A. True

B. False

Answer: A

Yes. The default Virtuoso Layout flow is sequential, and each task is carried out after completing the previous task.

In the CLE environment, you can create multiple partitions in the design and have designers work in parallel, which helps save time.

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Quiz: Concurrent Layout Modes

The commands that are enabled in the Concurrent menu depend on which of the modes in which the design is open?

Select all that apply.

A. Manager mode
B. Designer mode
C. Single user mode
D. Design partition mode
E. Peer designer mode
F. All of the above

Answer: A, B, and C

The commands that are enabled in the Concurrent menu depend on the mode in which the design is open. These modes are the Manager mode, the Designer mode, and the Single user mode.

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Quiz: Default Workspace for the Concurrent Layout

The default workspace for the Concurrent Layout is called ________.

Enter the answer in the blank field.

Answer: Concurrent_Layout

The default workspace for the Concurrent Layout is called Concurrent_Layout.

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Module 3

Initializing and Partitioning the Top Design in Manager Mode

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Module Objectives

In this module, you

- Access the Concurrent Layout assistant
- Initialize the top design for the Concurrent Layout editing
- Define an area-based design partition
- Define a layer-based design partition

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Accessing the Concurrent Layout Assistant

The *Concurrent Layout* assistant is a dockable assistant pane that provides various options to perform tasks related to concurrent layout editing. The assistant also displays alerts and other information about the design partition you are updating.

Use one of the following methods to access the *Concurrent Layout* assistant in the Layout canvas.

1. From the **Concurrent Menu**
2. From the **Assistants Menu**
3. From the **Layout XL/EXL/MXL Menu**
4. From the **Workspaces Menu**
5. From the **Workspaces Toolbar**

You can perform concurrent editing in Layout XL, Layout EXL, and Layout MXL.

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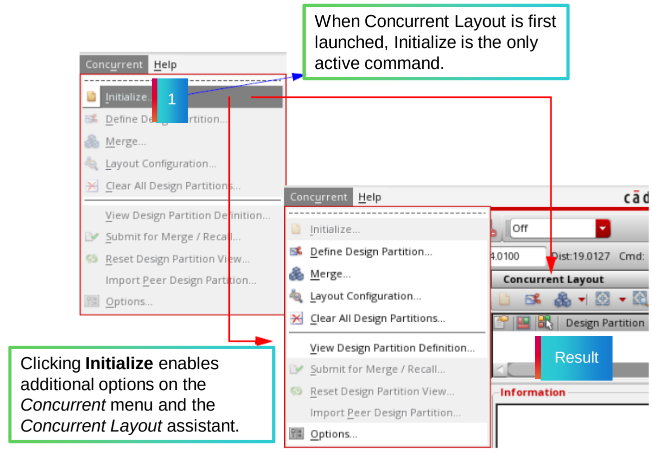
Method 1: Accessing the Concurrent Layout Assistant from the Concurrent Menu

To get the design ready for concurrent editing, choose the **Concurrent – Initialize** menu command from the menu bar.

Result: Displays additional commands in the *Concurrent* menu and embeds the *Concurrent Layout* assistant as a docked assistant pane within the current session window.



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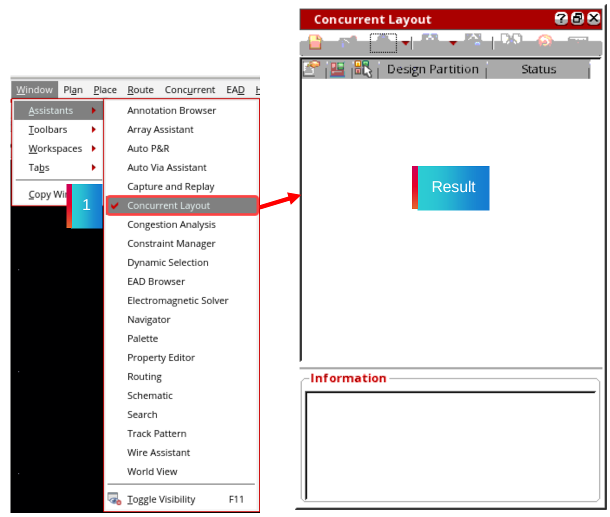
Method 2: Accessing the Concurrent Layout Assistant from the Assistants Menu

In the Layout XL/EXL/MXL window, choose **Window – Assistants – Concurrent Layout**.

Result: The *Concurrent Layout* assistant is enabled in the layout canvas.



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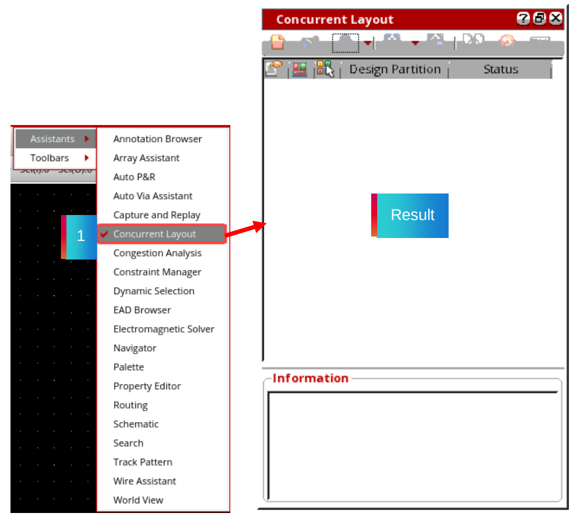
Method 3: Accessing the Concurrent Layout Assistant from the Layout XL/EXL/MXL Menu

Right-click the main Layout XL/EXL/MXL menu or toolbar area and select **Concurrent Layout**.

Result: The *Concurrent Layout* assistant is enabled in the layout canvas.



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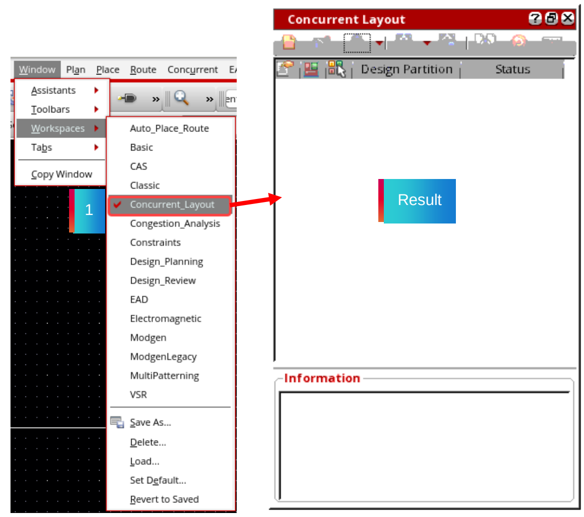
Method 4: Accessing the Concurrent Layout Assistant from the Workspaces Menu

Apply the Concurrent Layout workspace by selecting **Window – Workspaces – Concurrent_Layout**.

Result: The *Concurrent Layout* assistant is enabled in the layout canvas.



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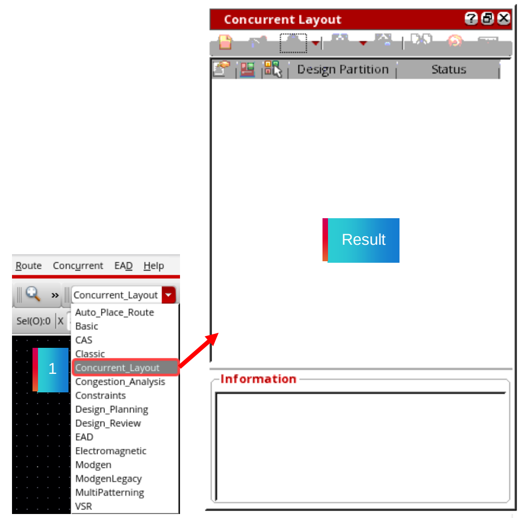
Method 5: Accessing the Concurrent Layout Assistant from the Workspaces Toolbar

Select **Concurrent_Layout** from the drop-down combo box on the *Workspace Configuration* toolbar.

Result: The *Concurrent Layout* assistant is enabled in the layout canvas.



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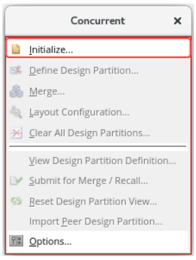
Initializing the Top Design for the Concurrent Layout Editing

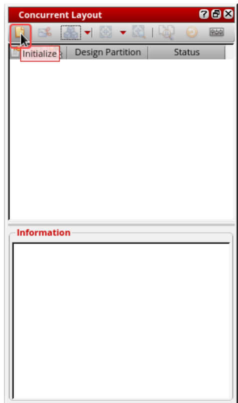
In the layout canvas, choose the **Concurrent – Initialize** menu command from the menu bar.

or

Click the **Initialize** icon on the *Concurrent Layout* assistant toolbar.

The first task for the manager is to initialize the design to get it ready for concurrent editing.





You get the Initialize pop-up window.

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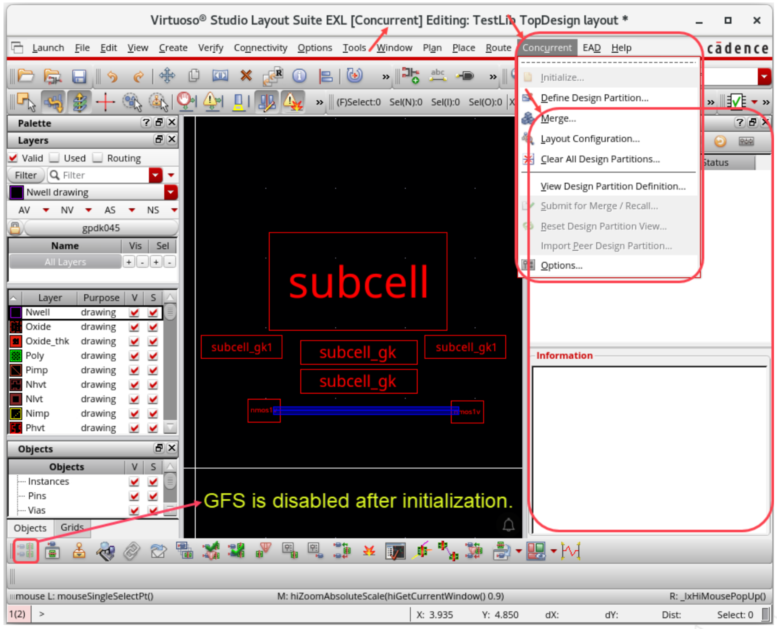
Initializing the Top Design for the Concurrent Layout Editing (continued)

The cellview is initialized for concurrent editing, and the Concurrent Layout assistant is displayed if it was not displayed earlier.



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Post Initialization Details

- Initialization saves some information about all the objects in the design. It is used as a reference later to identify and merge back the objects edited in a design partition.
- After the design is initialized for concurrent editing, additional options will be available on the *Concurrent* menu and the *Concurrent Layout* assistant.
- Layout configuration cellview *TestLib/TopDesign/layout_cle_layConfig* has been created.
- The cellview contains a text file that is automatically updated to list all subcells from top partitions that were incrementally edited during concurrent editing.

Generate All From Source is disabled after initialization because it can erase concurrent layout data saved in the initial design. Run this command before initialization.

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Defining an Area-Based Design Partition

After the design is initialized, you can start defining area-based design partitions based on your editing requirements.

You should partition the design carefully to avoid partition overlapping, which will increase the likelihood of edit conflicts among design partitions. Additionally, try not to use a global container, like a FigSet, across design partitions.

Refer to the steps given here for defining area-based design partitions.

You can define the following types of design partitions: Area-Based Design Partition, Layer-Based Design Partition, and Mixed Design Partition.

1. Invoke the Define Design Partition form.
2. Add **Area-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **New** button in the *Areas* field in the Define Design Partition form.
5. Click the **Add** button in the Define Design Partition form.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the button next to the **Split Crossing Objects** option to display.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to close the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

When a design is initialized for multiple users, you can start defining design partitions based on your editing requirements.

You can define the following types of design partitions:

Area-based design partition: One or more area partitions are attached to each design partition.

Layer-based design partition: A range of layers is specified for each design partition.

Mixed design partition: An area partition is specified along with a range of layers for a design partition.

You can specify all three types of design partitions at the same time based on your requirements.

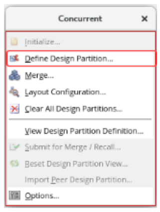
Step 1: Invoking the Define Design Partition Form

There are two methods to invoke the Define Design Partition form:

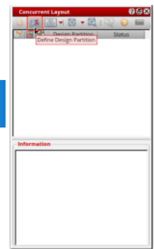
a. In the layout canvas, choose the **Concurrent – Define Design Partition** menu command from the menu bar.

b. Click the **Define Design Partition** button on the *Concurrent Layout* assistant.

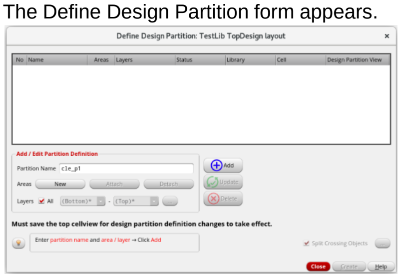
a



b



The Define Design Partition form appears.



1	Invoke the Define Design Partition form.
2	Add **Area-based design partition** in the Define Design Partition form.
3	Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4	Click the **New** button in the *Areas* field in the Define Design Partition form.
5	Click the **Add** button in the Define Design Partition form.
6	Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7	Click the button next to the **Split Crossing Objects** option to display.
8	Click the **Create** button after defining all the design partitions.
9	Click the **Close** button to close the Define Design Partition form.
10	Open the Library Manager to review the created design partition views and observe the changes.

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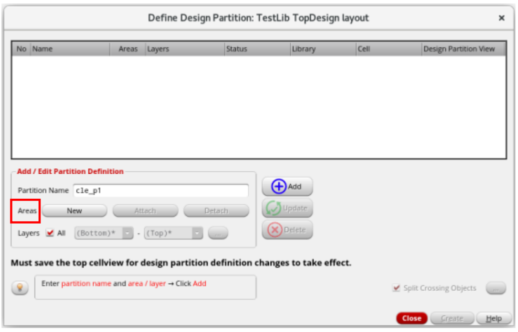
Step 2: Adding Area-Based Design Partition

Add **Area-based design partition** in the Define Design Partition form.

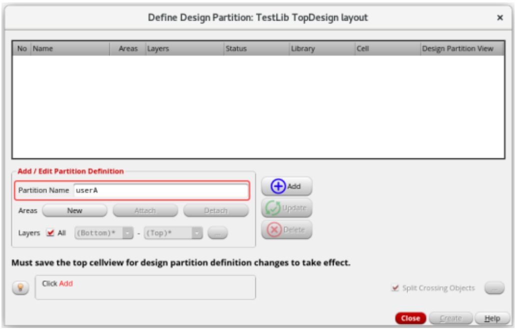


1	Invoke the Define Design Partition form.
2	Add **Area-based design partition** in the Define Design Partition form.
3	Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4	Click the **New** button in the *Areas* field in the Define Design Partition form.
5	Click the **Add** button in the Define Design Partition form.
6	Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7	Click the button next to the **Split Crossing Objects** option to display.
8	Click the **Create** button after defining all the design partitions.
9	Click the **Close** button to close the Define Design Partition form.
10	Open the Library Manager to review the created design partition views and observe the changes.

This page does not contain notes.

Step 3: Specifying the Name *userA* in the *Partition Name* Field

Specify a name, say, **userA**, in this example, for the new design partition in the *Partition Name* field in the Define Design Partition form.



The default name in this field is used if you do not specify a name. By default, the names of the new design partitions and the corresponding design partition views are **cle_px** and **layout_cle_px**, where **x** is a number.

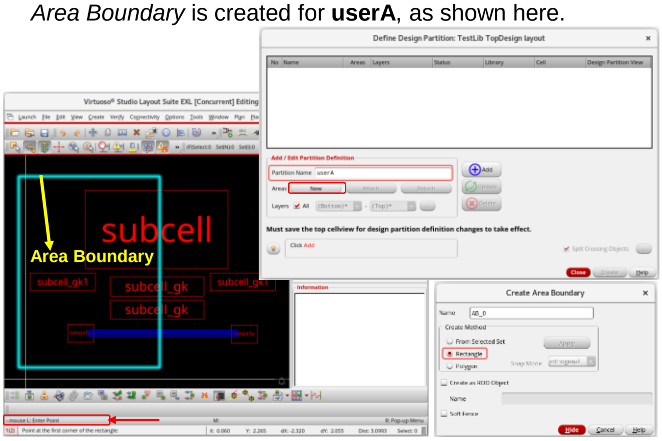
1. Invoke the Define Design Partition form.
2. Add **Area-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **New** button in the *Areas* field in the Define Design Partition form.
5. Click the **Add** button in the Define Design Partition form.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the button next to the **Split Crossing Objects** option to display.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to close the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

This page does not contain notes.

Step 4: Specifying the *Area Boundary* for the Design Partition *userA*

- Click the **New** button in the *Areas* field in the Define Design Partition form.
  - Press the **F3** key to display the *Create Area Boundary* form.
  - Select **Rectangle** as the *Create Method*.
  - Point at the *first corner* of the rectangle.
  - Point at the *opposite corner* of the rectangle.

Area Boundary is created for **userA**, as shown here.



1. Invoke the Define Design Partition form.
2. Add **Area-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **New** button in the *Areas* field in the Define Design Partition form.
5. Click the **Add** button in the Define Design Partition form.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the button next to the **Split Crossing Objects** option to display.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to close the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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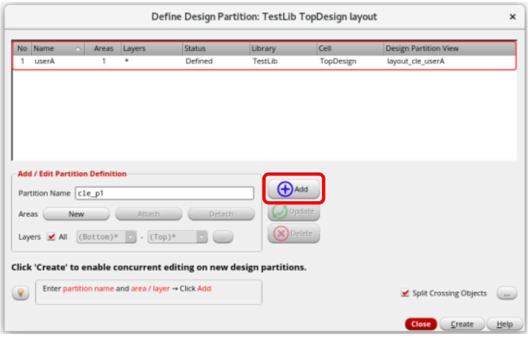
Step 5: Adding the Design Partition *userA*

Click the **Add** button in the Define Design Partition form.



A design partition named **userA** with corresponding design partition view, **layout_cle_userA**, with **1 Area** *Defined,* is added for the **TopDesign** cell in the **TestLib** library in the Define Design Partition form, as shown here.

The status of the newly added design partition, **userA**, is *Defined*. If a design partition view already exists, the status is *Reuse*.

1. Invoke the Define Design Partition form.
2. Add **Area-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **New** button in the *Areas* field in the Define Design Partition form.
5. Click the **Add** button in the Define Design Partition form.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the button next to the **Split Crossing Objects** option to display.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to close the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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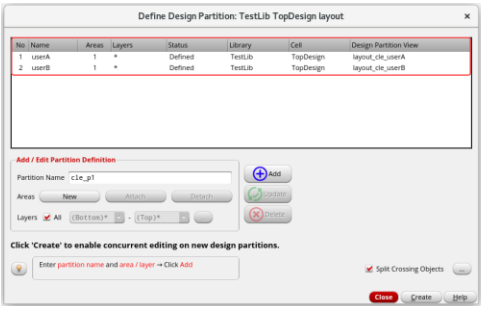
Step 6: Adding the Design Partition Definition for *userB*

Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.



You see the two design partitions, named **userA** and **userB**, with their corresponding design partition views, **layout_cle_userA**, and **layout_cle_userB**, with **1 Area** *Defined* each, are added for the **TopDesign** cell in the **TestLib** library in the Define Design Partition form, as shown here.

Use the *Attach* and *Detach* buttons on the Define Design Partition form to add or remove the area boundary associated with a design partition.
Use the *Update* and *Delete* buttons on the Define Design Partition form to update and delete the design partitions.

1. Invoke the Define Design Partition form.
2. Add **Area-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **New** button in the *Areas* field in the Define Design Partition form.
5. Click the **Add** button in the Define Design Partition form.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the button next to the **Split Crossing Objects** option to display.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to close the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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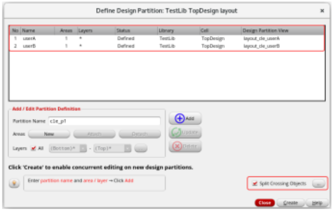
Step 7: Handling the Crossing Objects When Creating the Design Partitions

- Click the button next to the **Split Crossing Objects** option to display.



- The Split Crossing Objects Options form is displayed. Keep the default values and click **OK** in the form.

You can use the Split Crossing Objects Options form to change the settings of how crossing objects are handled when the design partitions are created.

1	Invoke the Define Design Partition form.
2	Add **Area-based design partition** in the Define Design Partition form.
3	Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4	Click the **New** button in the *Areas* field in the Define Design Partition form.
5	Click the **Add** button in the Define Design Partition form.
6	Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7	Click the button next to the **Split Crossing Objects** option to display.
8	Click the **Create** button after defining all the design partitions.
9	Click the **Close** button to close the Define Design Partition form.
10	Open the Library Manager to review the created design partition views and observe the changes.

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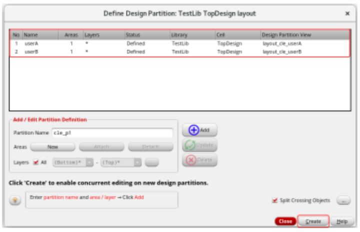
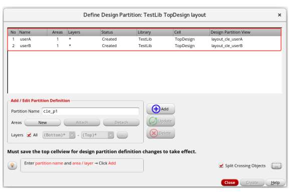
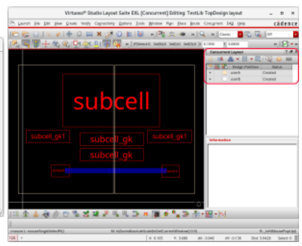
Step 8: Creating the Design Partition Views

- Click the **Create** button after you have completed defining all the design partitions to display the Create Design Partition View form.
- Click **OK** in this form.



Design partition views are created, and the status *Created* is displayed in the Define Design Partition form and the *Concurrent Layout* assistant.





1. Invoke the Define Design Partition form.
2. Add **Area-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **New** button in the *Areas* field in the Define Design Partition form.
5. Click the **Add** button in the Define Design Partition form.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the button next to the **Split Crossing Objects** option to display.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to close the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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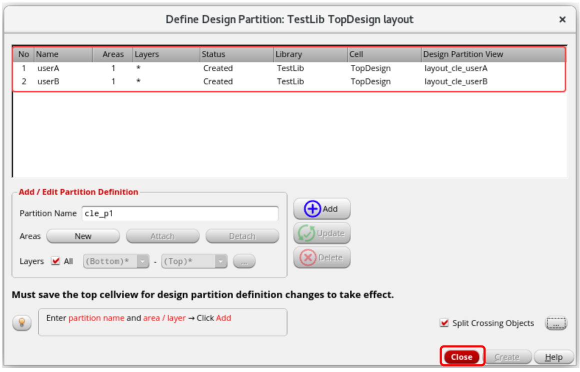
Step 9: Closing the Define Design Partition Form

Click the **Close** button to close the Define Design Partition form.



In the Design Management environment, an additional dialog box is displayed to confirm that new design partition views have been checked into the design management system.

You can only exit the Define Design Partition form by clicking the **Close** button.

1. Invoke the Define Design Partition form.
2. Add **Area-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **New** button in the *Areas* field in the Define Design Partition form.
5. Click the **Add** button in the Define Design Partition form.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the button next to the **Split Crossing Objects** option to display.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to close the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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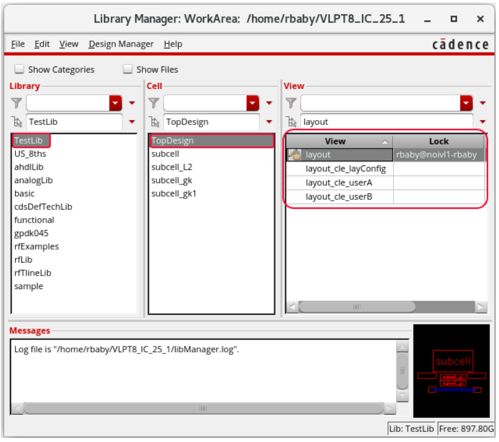
Step 10: Reviewing the Created Design Partition Views

Open the Library Manager to review the created design partition views and observe the changes.



TestLib/TopDesign/layout_cle_userA and *TestLib/TopDesign/layout_cle_userB* design partition views are created, as shown here.

1. Invoke the Define Design Partition form.
2. Add **Area-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **New** button in the *Areas* field in the Define Design Partition form.
5. Click the **Add** button in the Define Design Partition form.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the button next to the **Split Crossing Objects** option to display.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to close the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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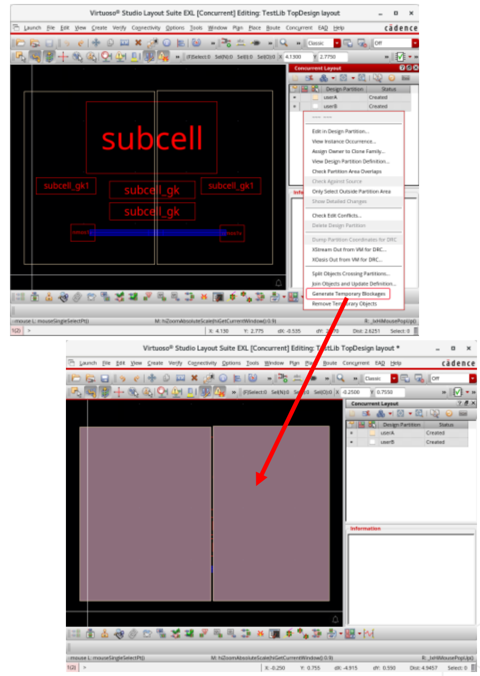
Generating the Temporary Blockages in Manager Mode

To generate and add the temporary blockages:

Right-click in the **Concurrent Layout** assistant – **Generate Temporary Blockages**.



- Temporary blockages are added at the area boundaries.
- You need to generate temporary blockages at the area boundaries of the design partitions for space-based routing between Concurrent Layout design partitions.
- These blockages are needed because the space-based router connects to the temporary Concurrent Layout pins only at the area boundary.
  - No routing takes place within the design partitions.

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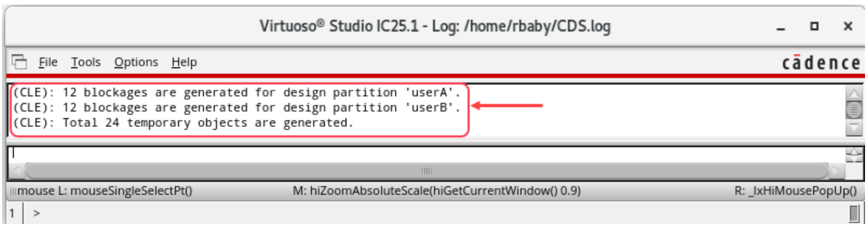
Generated Objects Information in the CIW

View the CIW for the temporary objects generated.



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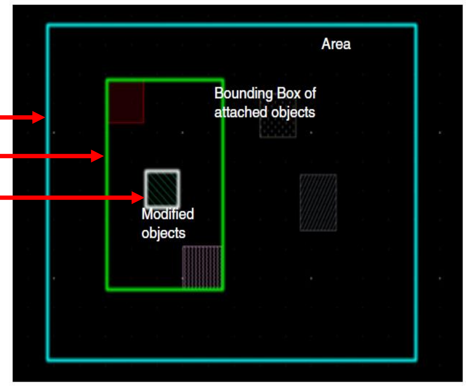
Using Colors to Identify Objects in an Area-Based Design Partition

Designers may accidentally edit and modify the content in their design partition if they do not use colors in it.

- Use colors to identify the objects in a selected design partition.
- It helps to prevent the users from touching the modified objects while using the *Attach* and *Detach* commands.

Object	Color
Area	Light blue
The bounding box of attached objects	Green
Modified objects	White

An example to show the different colors used to highlight the modified objects.



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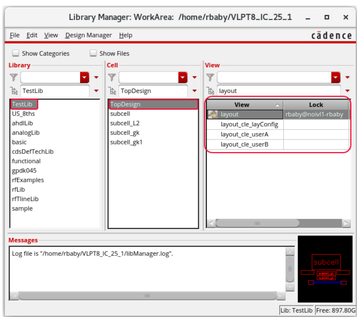
Defining a Layer-Based Design Partition

After the design is initialized, you can start defining layer-based design partitions based on your editing requirements.

The created design partition views are of minimum size. This is because only the changes made to the associated design partitions are saved in this view.



Refer to the steps given here for defining layer-based design partitions.

1. Invoke the Define Design Partition form.
2. Add **Layer-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **Add** button in the Define Design Partition form.
5. Specify the top and bottom layers to be included in the **Layers** field.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the **Layer Preview** button to add/remove the layers.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to exit the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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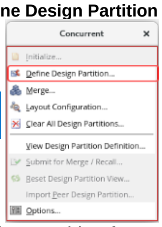
Step 1: Invoking the Define Design Partition Form

There are two methods to invoke the Define Design Partition form:

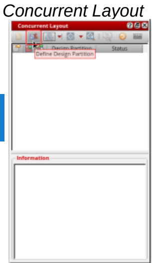
a. In the layout canvas, choose the **Concurrent – Define Design Partition** menu command from the menu bar.

b. Click the **Define Design Partition** button on the *Concurrent Layout* assistant.

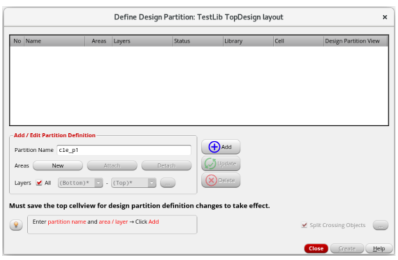
a



b



The Define Design Partition form appears.



1. Invoke the Define Design Partition form.
2. Add **Layer-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **Add** button in the Define Design Partition form.
5. Specify the top and bottom layers to be included in the **Layers** field.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the **Layer Preview** button to add/remove the layers.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to exit the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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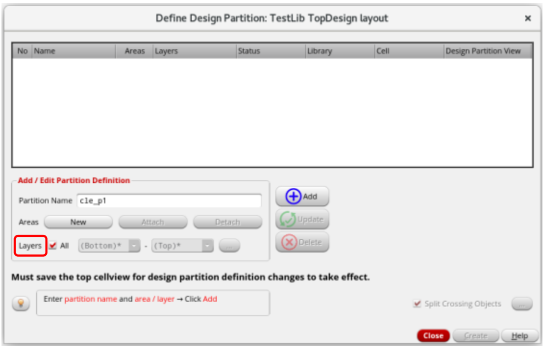
Step 2: Adding Layer-Based Design Partition

Add **Layer-based design partition** in the Define Design Partition form.

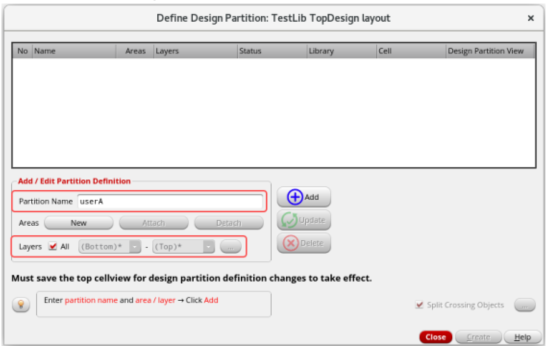


1. Invoke the Define Design Partition form.
2. Add **Layer-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **Add** button in the Define Design Partition form.
5. Specify the top and bottom layers to be included in the **Layers** field.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the **Layer Preview** button to add/remove the layers.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to exit the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

This page does not contain notes.

Step 3: Specifying the Name *userA* in the *Partition Name* Field

Specify a name, say, **userA**, in this example, for the new design partition in the *Partition Name* field in the Define Design Partition form.



The default name in this field is used if you do not specify a name. By default, the names of the new design partitions and the corresponding design partition views are **cle_px** and **layout_cle_px**, where **x** is a number.

1. Invoke the Define Design Partition form.
2. Add **Layer-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **Add** button in the Define Design Partition form.
5. Specify the top and bottom layers to be included in the **Layers** field.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the **Layer Preview** button to add/remove the layers.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to exit the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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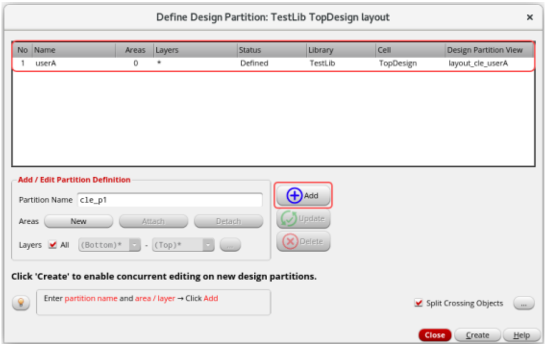
Step 4: Adding the Design Partition *userA*

Click the **Add** button in the Define Design Partition form.



A design partition named **userA**, with corresponding design partition view, **layout_cle_userA**, with *** Layers** *Defined* added for the **TopDesign** cell in **TestLib** library in the Define Design Partition form, as shown here.

The status of the newly added design partition, **userA**, is *Defined*. If a design partition view already exists, the status is *Reuse*.

1	Invoke the Define Design Partition form.
2	Add **Layer-based design partition** in the Define Design Partition form.
3	Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4	Click the **Add** button in the Define Design Partition form.
5	Specify the top and bottom layers to be included in the **Layers** field.
6	Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7	Click the **Layer Preview** button to add/remove the layers.
8	Click the **Create** button after defining all the design partitions.
9	Click the **Close** button to exit the Define Design Partition form.
10	Open the Library Manager to review the created design partition views and observe the changes.

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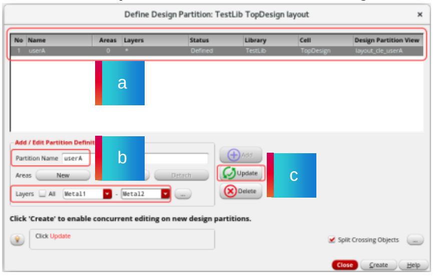
Step 5: Specifying the Top and Bottom Layers in the *Layers* Field

a. Select the design partition **userA**.

b. Specify the top and bottom layers to be included in the design partition in the **Layers** field.

c. Click the **Update** button in the Define Design Partition form.



The layers **Metal1 - Metal2(3)** are included in the **Layers** field in the Define Design Partition form.

1. Invoke the Define Design Partition form.
2. Add **Layer-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **Add** button in the Define Design Partition form.
5. Specify the top and bottom layers to be included in the **Layers** field.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the **Layer Preview** button to add/remove the layers.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to exit the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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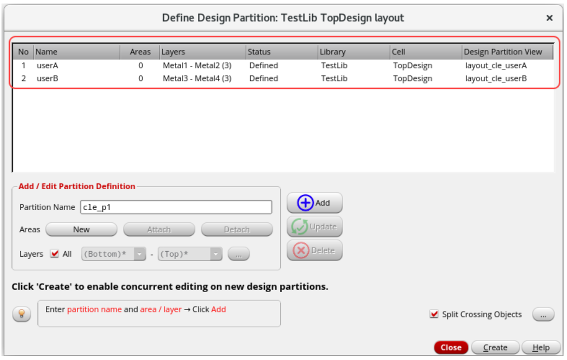
Step 6: Adding the Design Partition Definition for *userB*

Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.



You see the two design partitions, named **userA** and **userB**, with their corresponding design partition views, **layout_cle_userA**, and **layout_cle_userB**, with **Metal1 - Metal2(3) and Metal3 - Metal4(3) Layers** *Defined* are added for the **TopDesign** cell in **TestLib** library in the Define Design Partition form as shown here.

1	Invoke the Define Design Partition form.
2	Add **Layer-based design partition** in the Define Design Partition form.
3	Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4	Click the **Add** button in the Define Design Partition form.
5	Specify the top and bottom layers to be included in the **Layers** field.
6	Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7	Click the **Layer Preview** button to add/remove the layers.
8	Click the **Create** button after defining all the design partitions.
9	Click the **Close** button to exit the Define Design Partition form.
10	Open the Library Manager to review the created design partition views and observe the changes.

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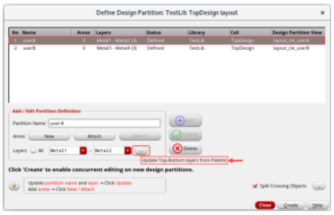
Step 7: Using the *Layer Preview* Button to Add/Remove the Layers in the Define Design Partition Form

- (Optional) Select the design partition, **userA**, and click the **Update Top-Bottom layers from Palette** (**Layer Preview**) button in the **Layers** field in the Define Design Partition form to remove certain layers from the layer range or add other visible layers from the Palette.



The *Partition Layers* form for **userA** appears. Use this form to:

- Preview the layers included in the design partition.
- Clear the check boxes of the layers you do not want to include in the design partition.
- Click **Import Visible from Palette** to import additional layers from the Palette.

1	Invoke the Define Design Partition form.
2	Add **Layer-based design partition** in the Define Design Partition form.
3	Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4	Click the **Add** button in the Define Design Partition form.
5	Specify the top and bottom layers to be included in the **Layers** field.
6	Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7	Click the **Layer Preview** button to add/remove the layers.
8	Click the **Create** button after defining all the design partitions.
9	Click the **Close** button to exit the Define Design Partition form.
10	Open the Library Manager to review the created design partition views and observe the changes.

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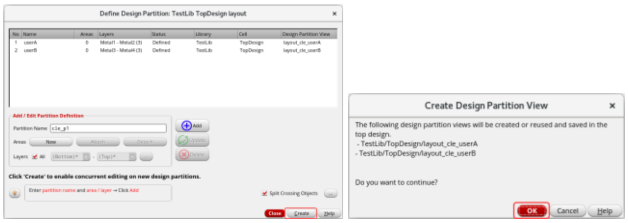
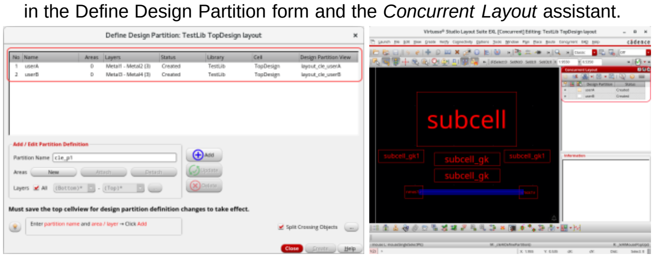
Step 8: Creating the Design Partition Views

- Click the **Create** button after you have completed defining all the design partitions to display the Create Design Partition View form.
- Click **OK** in this form.



Design partition views are created, and the status *Created* is displayed in the Define Design Partition form and the *Concurrent Layout* assistant.



1. Invoke the Define Design Partition form.
2. Add **Layer-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **Add** button in the Define Design Partition form.
5. Specify the top and bottom layers to be included in the **Layers** field.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the **Layer Preview** button to add/remove the layers.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to exit the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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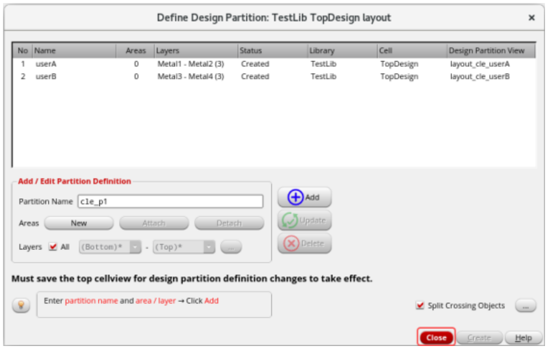
Step 9: Closing the Define Design Partition Form

Click the **Close** button to exit the Define Design Partition form.



In the Design Management environment, an additional dialog box is displayed to confirm that new design partition views have been checked into the design management system.

You can only exit the Define Design Partition form by clicking the **Close** button.

1. Invoke the Define Design Partition form.
2. Add **Layer-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **Add** button in the Define Design Partition form.
5. Specify the top and bottom layers to be included in the **Layers** field.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the **Layer Preview** button to add/remove the layers.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to exit the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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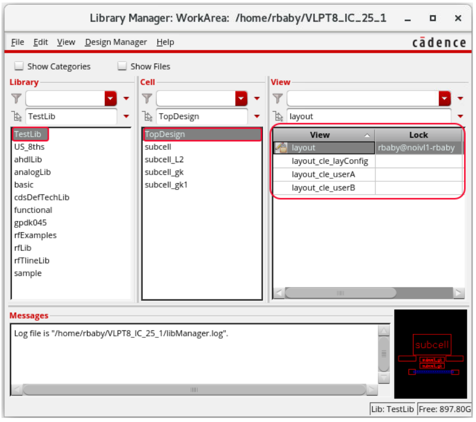
Step 10: Reviewing the Created Design Partition Views

Open the Library Manager to review the created design partition views and observe the changes.



TestLib/TopDesign/layout_cle_userA and *TestLib/TopDesign/layout_cle_userB* design partition views are created, as shown here.

1. Invoke the Define Design Partition form.
2. Add **Layer-based design partition** in the Define Design Partition form.
3. Specify **userA** in the *Partition Name* field in the Define Design Partition form.
4. Click the **Add** button in the Define Design Partition form.
5. Specify the top and bottom layers to be included in the **Layers** field.
6. Use the steps shown in **Steps 2-5** and add the design partition definition for **userB**.
7. Click the **Layer Preview** button to add/remove the layers.
8. Click the **Create** button after defining all the design partitions.
9. Click the **Close** button to exit the Define Design Partition form.
10. Open the Library Manager to review the created design partition views and observe the changes.

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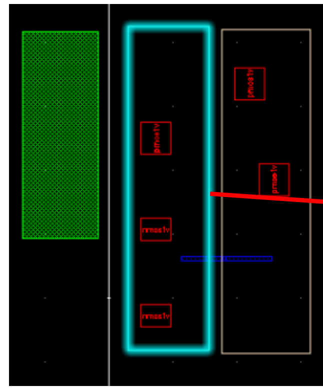
Edit Scope of a Design Partition

Edit Scope for each design partition is set to select only those objects that are inside the current partition by default.

- The Concurrent Layout uses the definition you made in the Define Design Partition form to identify which object belongs to which design partition.
- Information about the objects undergoing modification in a design partition is stored in the associated design partition view.



Halo around the area boundary.

If **Edit Scope** is set to *Off*, editing objects outside the partition generates alerts in the *Concurrent Layout* assistant.
You will have to sign off on each alert and warning individually.

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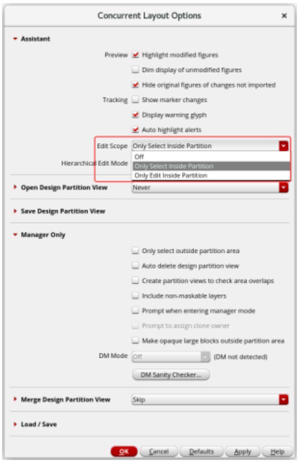
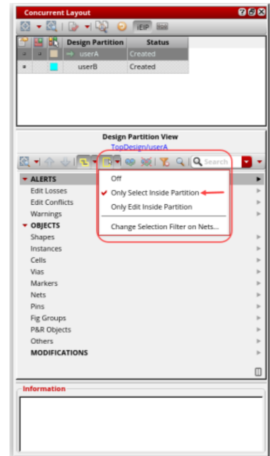
Edit Scope of a Design Partition (continued)

You can change the **Edit Scope** settings using the *Concurrent Layout Options* form in designer mode.



You can change the **Edit Scope** settings using the *Concurrent Layout* assistant in designer mode.



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Commands to Be Used to Grant the Write Permission to Open a Design Partition View

Designers must have write permission to open a design partition view for concurrent editing.

- In the Design Management (DM) environment, file permission is handled by the DM check-in and check-out process.
- Otherwise, the manager can use UNIX commands to change the file permissions.

When the manager and designers are in the same group, the following command can be used to grant the group write permission:

- *chmod -R g +w <design_partition_views>*
- Alternatively, you can change **UMASK** as shown here before launching Virtuoso to set the file permission for all the files created by the current Virtuoso process.
  - *% umask => 22*

- Change the **UMASK** setting and start Virtuoso.
  - You might need this setting when the manager and designers are not in the same group.
    - *% umask 0*
    - *% virtuoso &*
- After you have defined and saved new design partitions, exit Virtuoso and restore the **UMASK** setting by using the following command:
  - *% umask 22*

The **UMASK** change affects all new files in the current Virtuoso session.

If you use commands such as **Save As** and **Copy**, the file permissions might not be as expected.

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Module Summary

In this module, you learned how to

- Access the Concurrent Layout assistant
- Initialize the top design for the Concurrent Layout editing
- Define an area-based design partition
- Define a layer-based design partition

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Labs

Lab 3-1 Initializing the Top Design in Manager Mode

Lab 3-2 Partitioning the Top Design in Manager Mode

This page does not contain notes.

Quiz: Defining the Design Partition Types

What are the ways can you use to define the design partition types?

Select all that apply.

A. Area-Based Design Partition
B. Layer-Based Design Partition
C. Mixed Design Partition
D. Width-Based Design Partition
E. Length-Based Design Partition
F. All of the above

Answer: A, B, and C

You can define the area-based, layer-based, and mixed design partition types.

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Quiz: Space-Based Routing Between Concurrent Layout Design Partitions

You need to generate temporary _________ at the area boundaries of the design partitions for space-based routing between Concurrent Layout design partitions.

Enter the answer in the blank field.

Answer: blockages

You need to generate temporary blockages at the area boundaries of the design partitions for space-based routing between the Concurrent Layout design partitions.

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Quiz: Identifying the Objects in a Selected Design Partition

You can use _______ to identify the objects in a selected design partition.

Enter the answer in the blank field.

Answer: colors

You can use colors to identify the objects in a selected design partition.

This page does not contain notes.

Quiz: Updating the Layers in the Define Design Partition Form

You can use the Update Top-Bottom layers from Palette button in the Layers field to add/remove the layers in the Define Design Partition form.

A. True

B. False

Answer: A

Yes. You can use the Update Top-Bottom layers from Palette button in the Layers field to add/remove the layers in the Define Design Partition form.

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Quiz: Changing the Edit Scope of a Design Partition

Edit Scope of a design partition can be changed in _________ mode.

Enter the answer in the blank field.

Answer: designer

The Edit Scope of a design partition can be changed in designer mode.

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Module 4

Editing the Design Partition userA in Designer Mode

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Module Objectives

In this module, you

- Set the Edit Scope for the design partition userA in designer mode
- Edit the design partition userA using the CLE flow
- Edit the design partition userA hierarchically using the CLE flow
- Verify the incremental EIP updates
  - subcell/cle_userA_TopDesign
- Submit the incremental EIP updates for merge
  - TopDesign/layout_cle_userA and subcell/layout_cle_userA_TopDesign
- Perform automatic routing in:
  - An area-based design partition
  - A layer-based design partition

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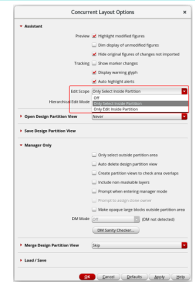
Setting the Edit Scope for the Design Partition userA in Designer Mode

The **Edit Scope** for each design partition is set to select only those objects that are inside the current partition by default.

You can change these settings using one of the following two methods in designer mode:

1. Setting the Edit Scope for **userA** using the *Concurrent Layout Options* form.
2. Setting the Edit Scope for **userA** using the *Concurrent Layout* assistant.



The following options can be used to configure the edit scope for a designer based on requirements:

Off	Disables the edit mode settings.
Only Select Inside Partition	Lets you select only those objects that are inside the current partition. Objects that are fully or partially inside or have been temporarily moved outside the current partition for editing are also selectable.
Only Edit Inside Partition	Lets you edit only those objects that are inside the current partition. When you specify this option, all edit commands are restricted to the specified design partition. You cannot edit objects outside this design partition.

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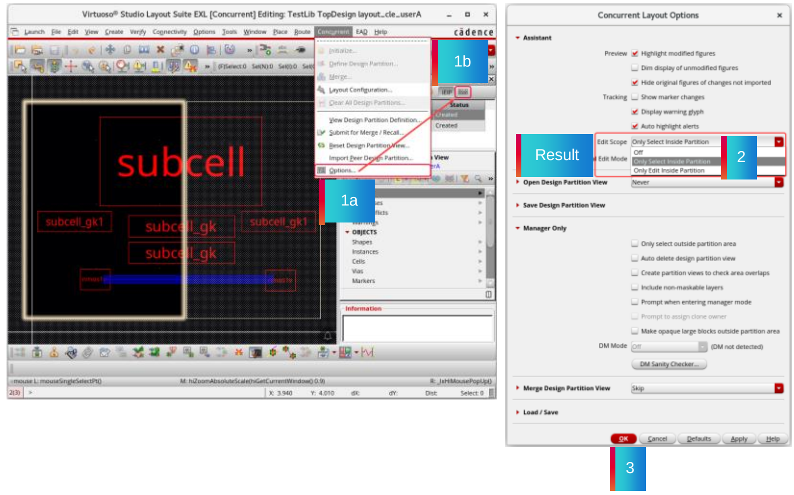
Method 1: Setting the Edit Scope for userA Using the Concurrent Layout Options Form

To set the edit scope for the **userA**, do the following steps:

1. Open the *Concurrent Layout Options* form using any one of the two methods:
   a. From the layout window, choose the **Concurrent – Options** menu command.
   b. Click the **Options** icon in the *Concurrent Layout* assistant to display the *Concurrent Layout Options* form.
2. In the *Concurrent Layout Options* form, in the *Edit Scope* cyclic field, choose the required option. In this case, select the default option, *Only Select Inside Partition*.
3. Click **OK** in the form.

Result: *Edit Scope* is set to *Only Select Inside Partition*.



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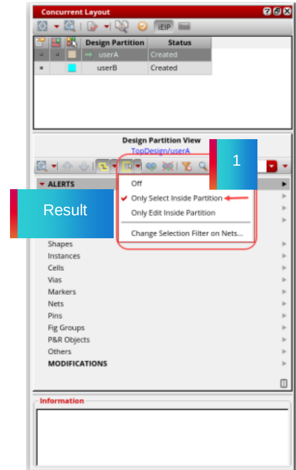
Method 2: Setting the Edit Scope for userA Using the Concurrent Layout Assistant

To set the edit scope for **userA**, perform the following steps:

1. In the *Concurrent Layout* assistant, from the *Edit Scope* cyclic field, choose the required option. In this case, select the default option, *Only Select Inside Partition*.

Result: *Edit Scope* is set to *Only Select Inside Partition*.



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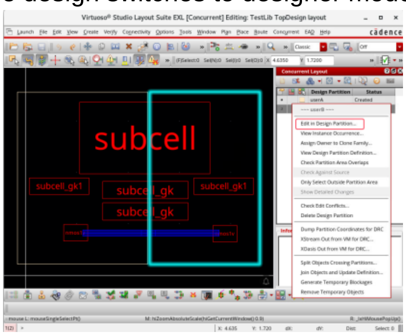
Concurrent Layout: Single-User and Multiple-User Modes

When one user is working in the Concurrent Layout, you can follow the steps below to enable designer mode for the user.

When a *single user* is editing the design in the *Concurrent Layout* assistant:

- **Right**-click the design partition you want to edit and select *Edit in Design Partition*.

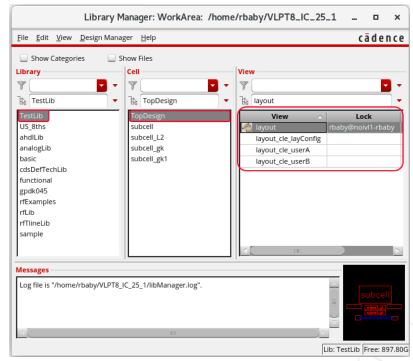
The design switches to designer mode.



When many users are working in the Concurrent Layout, you can follow the steps below to enable designer mode for them.

When *multiple users* are working on different design partitions:

- Open the Library Manager and then select and open the design partition view you want to edit.



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Best Practices: Concurrent Layout Editing

There are certain updates that should be made only in the top design.

Avoid making these updates in design partitions.

- Net deletion and property changes of cellviews (cv~>prop) are not merged or imported.
  - Net deletions can trigger unexpected deletions in the current design partition.
  - They are automatically updated in the top design.
  - Similarly, property changes can cause edit conflicts, so they are updated only in the top design.
- Snap Boundary and P&R Boundary are global and should be modified only in the top design.

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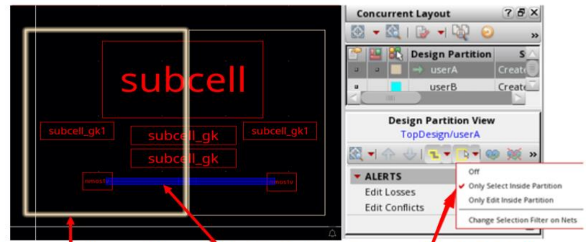
Features: Editing in a Design Partition

In designer mode, by default, in area-based design partitions, the area boundary for the selected design partition is highlighted, and the region outside it is grayed out.

- This happens because the *Only Select Inside Partition* option is enabled by default in the *Concurrent Layout* assistant.
- It helps in preventing the users from selecting any objects outside their area partition.
- Additionally, any wire crossing the design partition is split by default at the boundary.



Halo around the area partition highlighting the edit scope.

Wires crossing the design partition are split at the boundary.

The default edit scope is set.

Editing in a layer-based design partition is similar to editing in an area-based design partition.

The difference is that your edit scope is set to the range of layers included in that design partition.

You can also perform tasks such as automatic routing and interactive wire editing in these design partitions.

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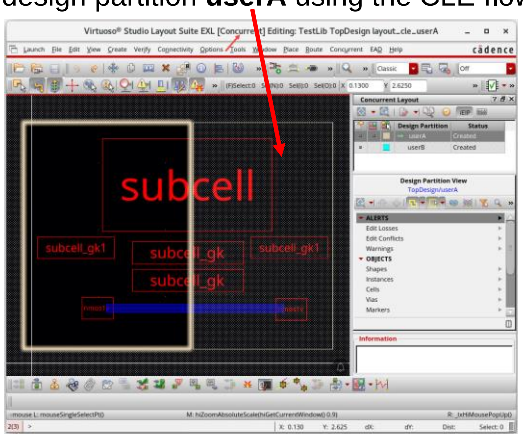
Editing the Design Partition userA Using the CLE Flow

After the required design partition and design partition views are created, designers can start editing the allocated design partition views. Designers should have write permission on the design partition views that store the edits of the top design.

You will explore editing the design partition **userA** using the CLE flow:

1. Editing Inside **userA**
2. Editing Outside **userA**



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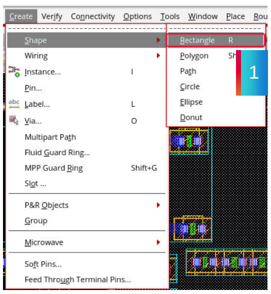
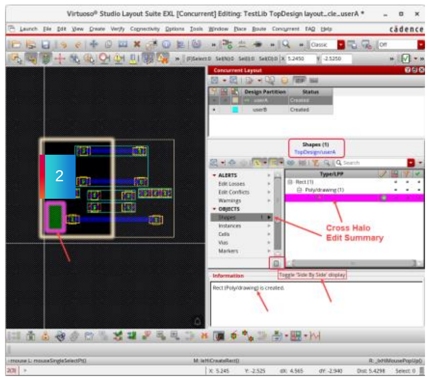
Editing Inside the Design Partition userA Using the CLE Flow

You will make some edits inside the design partition **userA**.

1. As an example, in the layout canvas, choose the **Create – Shape – Rectangle** menu command or press the bindkey **R** to invoke the Create Rectangle command.
2. Create a rectangle inside the design partition, as shown here.





Result

Observe the *Edit Summary* and the *Cross Halo* results for the rectangle shape created inside the design partition **userA** in the *Concurrent Layout* assistant.

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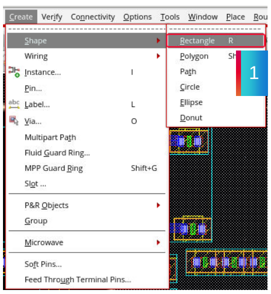
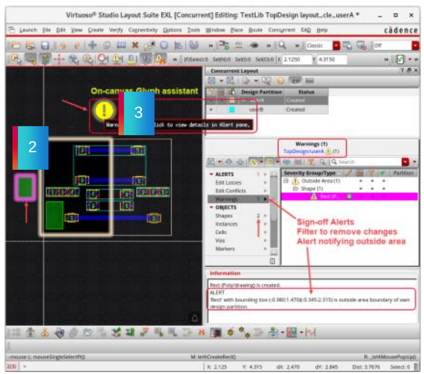
Editing Outside the Design Partition userA Using the CLE Flow

You will make some edits outside the design partition **userA**.

1. As an example, in the layout canvas, choose the **Create – Shape – Rectangle** menu command or press the bindkey **R** to invoke the Create Rectangle command.
2. Create a rectangle outside the design partition, as shown here.
3. Click the **On-canvas Glyph assistant** to view the details in the *Alert* pane.





Result

Observe the *Sign-off Alerts* and the *Alert notifying outside area* result for the rectangle shape created outside the design partition **userA** in the *Concurrent Layout* assistant.

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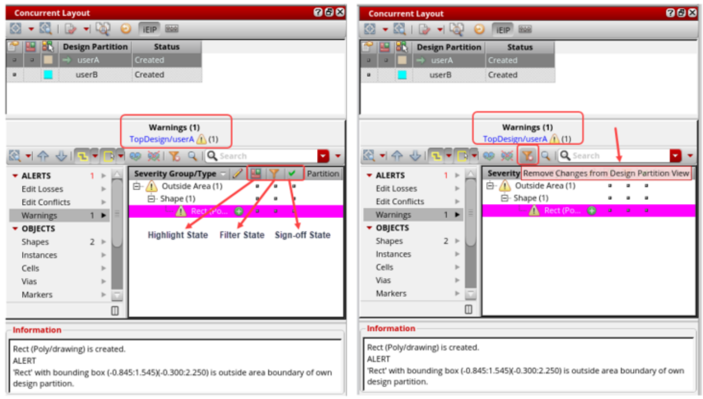
Options in the Concurrent Layout Assistant to Remove the Warnings

Observe the different options in the *Concurrent Layout* assistant.



Outside area edits need to be signed off on to indicate if these edits are to be retained and were done intentionally. However, if any edits are not to be retained, click in the ***Filter State*** column to first mark these changes in the assistant and then use the **Remove Changes from Design Partition View** button to remove them.

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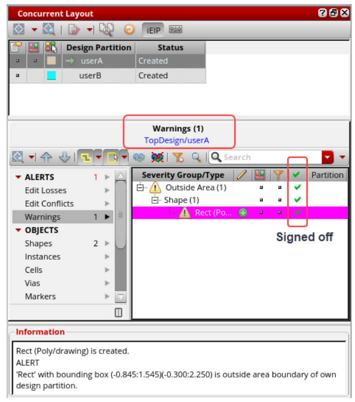
Signing Off the Outside Area Edits

Click the **Sign-off State** column to remove the warning and sign off the outside area edits alert.



The outside area edit is signed off, and the alert is removed.

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Editing the Design Partition userA Hierarchically Using the CLE Flow

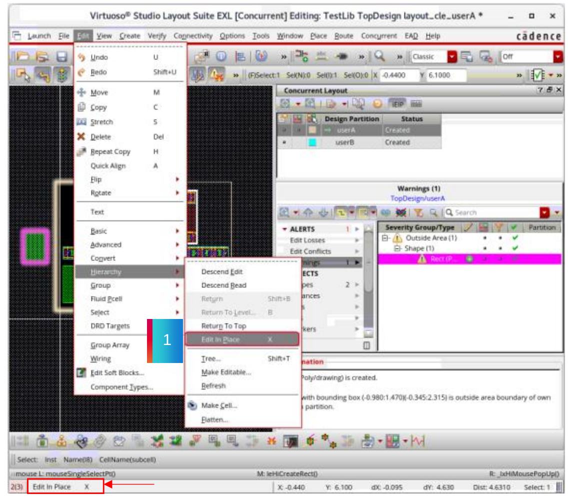
Hierarchical editing in the CLE flow lets multiple designers concurrently Edit In Place (EIP) into hierarchical subcells.

You will make hierarchical edits to the design partition **userA** using the CLE flow. You will descend into the instance *subcell* and do hierarchical editing on it.

1. First, in the layout canvas,
   - Select the instance *subcell* and choose the **Edit – Hierarchy – Edit In Place** menu command.

   or

   - Press the bindkey **X** to EIP to display the Hierarchical Edit Setup form.
2. Then, in the Hierarchical Edit Setup form, with the default options,
   - Click **OK** to create the hierarchical design partition views for the subcell.
3. Next, create a shape in the hierarchical design partition view *layout_cle_userA_TopDesign*.



Result: You have edited the hierarchical instance *subcell* using the CLE flow.

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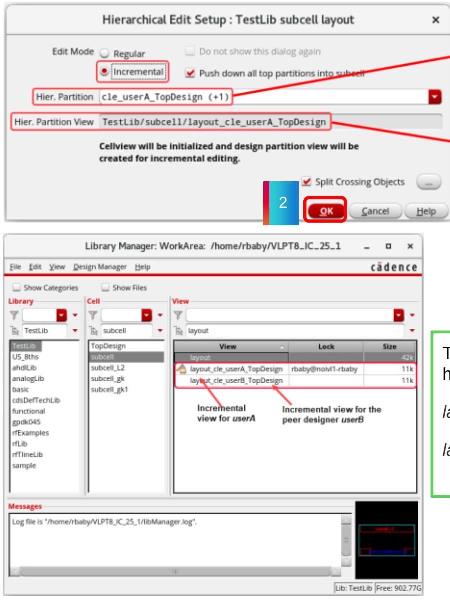
Result: Editing the Design Partition userA Hierarchically Using the CLE Flow



Hierarchical design partition views are created for current and peer designers.

<layout>
<top_design_partition_view>
<top_design_name>

Two incremental hierarchical design partition views are added for the hierarchical instance *subcell*, as seen in the Library Manager.

layout_cle_userA_TopDesign is the incremental view for *userA*.

layout_cle_userB_TopDesign is the incremental view for *userB*.

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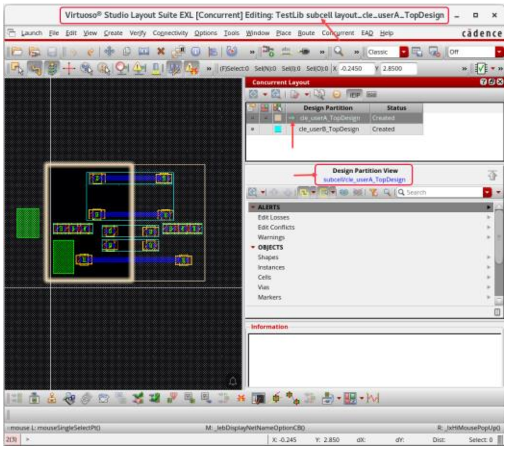
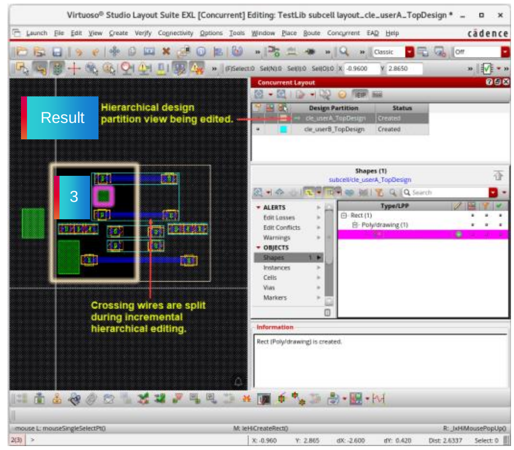
Result: Editing the Design Partition userA Hierarchically Using the CLE Flow (continued)

You are now descended into *TestLib/subcell/layout_cle_userA_TopDesign* cellview.



You have edited the hierarchical instance *subcell* using the CLE flow.



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Returning to the Top Partition Cellview: TestLib/TopDesign/layout_cle_userA

After the hierarchical editing (EIP) in the CLE flow is completed, designers will return to the top partition cellview.

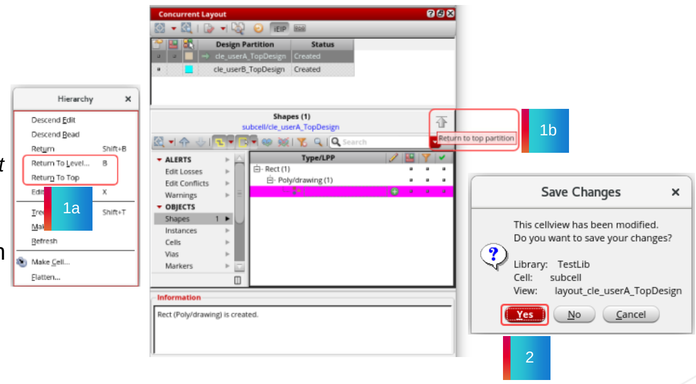
To return to the top partition cellview:

1. First, select any one of the two methods:
   a. In the layout canvas, choose the **Edit – Hierarchy – Return To Level (**bindkey **B**) / **Return To Top** menu command.
   b. Click the **Return to top partition** button in the *Concurrent Layout* assistant.

   You get the *Save Changes* form to inform you to save the changes you have made in the hierarchical design partition view *layout_cle_userA_TopDesign*.

2. In the *Save Changes* form, click the **Yes** button to save the changes in the hierarchical design partition view *layout_cle_userA_TopDesign*.

Result: You are now returned to the top design partition view *TopDesign/userA*.



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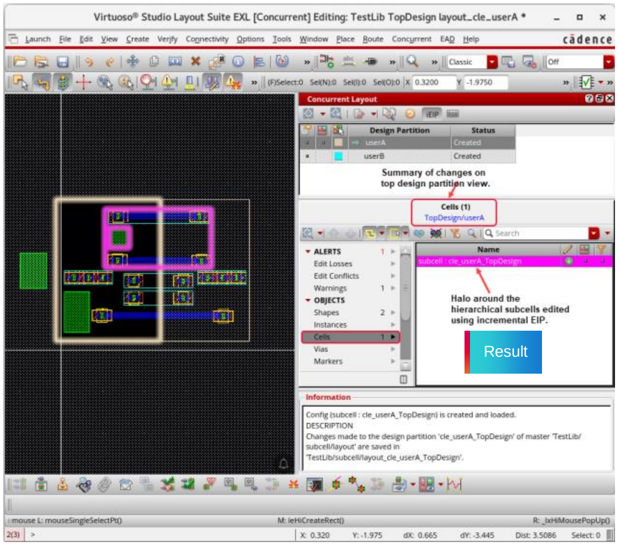
Result: Returning to the Top Partition Cellview:
TestLib/TopDesign/layout_cle_userA



CLE assistant summarizes all the edits performed in the top design partition view (in this case, *TopDesign/userA*) and incrementally edited hierarchical subcells (in this case, *subcell/cle_userA_TopDesign*. The hierarchical subcells edited in regular mode are excluded.

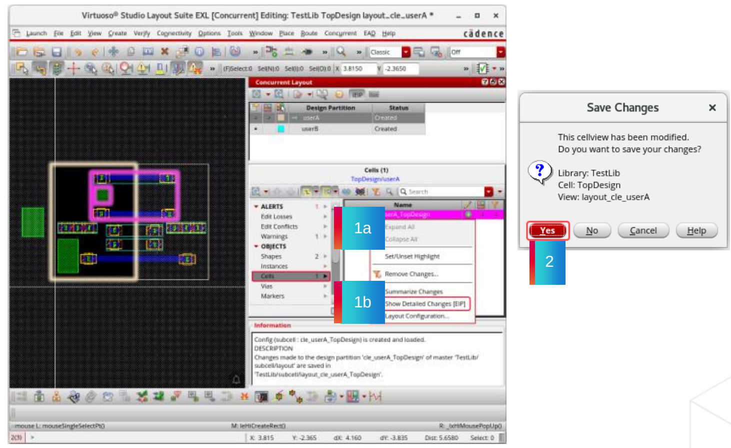
This page does not contain notes.

Verifying the Incremental EIP Updates in *subcell/cle_userA_TopDesign*

After you have completed the hierarchical editing, you can view and verify the incremental Edit In Place updates in the hierarchical instance subcell in the design partition **userA** using the **Show Detailed Changes (EIP)** option.

1. To view and verify the details of the edit changes inside the incrementally edited hierarchical subcell:
   a. In the layout canvas, **right**-click the object representing the hierarchical design partition view *subcell:cle_userA_TopDesign*.
   b. Choose the **Show Detailed Changes (EIP)** option.
2. Save the changes in the top design partition view *TopDesign/userA* by clicking the **Yes** button.
3. On the *Concurrent Layout* assistant, expand the correct category to view the edit changes saved in the hierarchical design partition view *subcell/cle_userA_TopDesign*.

Result: You have viewed and verified the incremental EIP updates in *subcell/cle_userA_TopDesign*.



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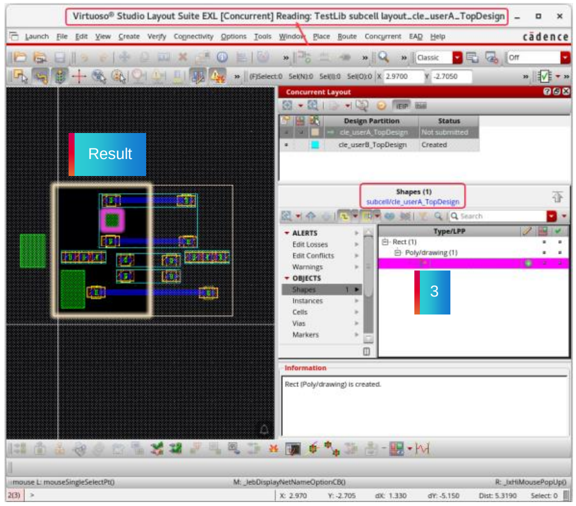
Result: Verifying the Incremental EIP Updates in *subcell/cle_userA_TopDesign*

When you choose **Show Detailed Changes [EIP]**, EIP is done into the hierarchical design partition view *subcell/cle_userA_TopDesign* in read-mode.



EIP will be performed in read-only mode. The object is shown in the *Cells* category in the *OBJECTS* section of the *Summary* Pane.

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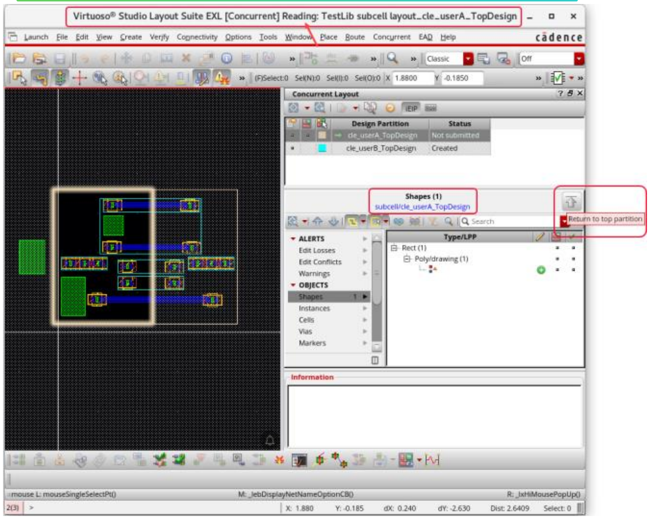
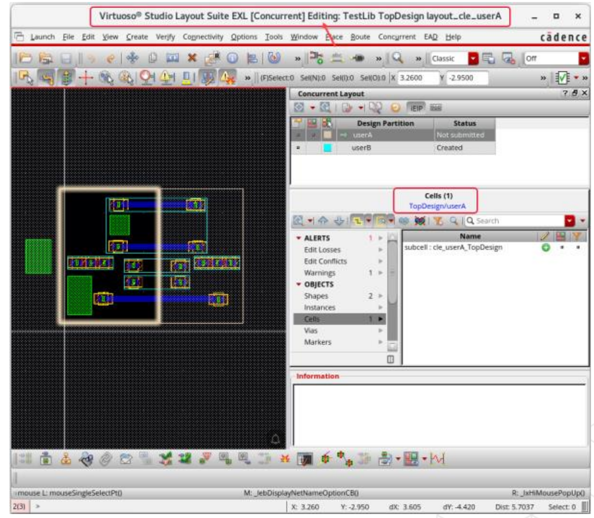
Returning to the Top Design Partition View *TopDesign/userA*

After reviewing all the changes, click the **Return to top partition** button on the *Concurrent Layout* assistant to return to the top design partition view.



Now you are returned to the top design partition view *TopDesign/userA*.



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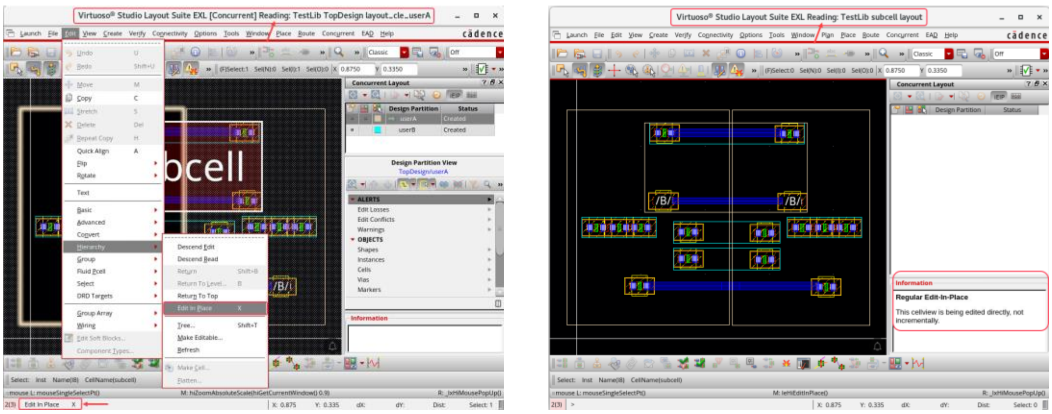
Behavior of Incremental EIP from a Read-Only Design Partition

Incremental EIP cannot be done from a read-only top design partition.

When you EIP from a read-only top design partition or if a subcell cannot be opened for editing, EIP is done only in ***Regular*** mode.



The **Information** pane in the Concurrent Layout assistant is used to inform you that you are editing in Regular mode.

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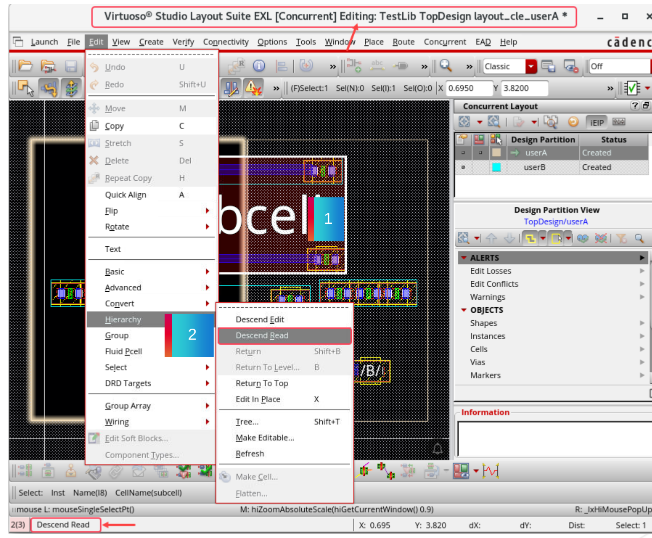
Editing a Subcell Incrementally After Descending in Read Mode

You can edit a hierarchical subcell in *Regular* or *Incremental* mode after you descend in read mode.

To edit a hierarchical subcell in *Incremental* mode:

1. In the Layout XL or Layout EXL canvas, select the instance *subcell.*
2. Choose the **Edit – Hierarchy – Descend Read** menu command to descend the hierarchical subcell in read mode.
3. In the hierarchical subcell, choose the **File – Make Editable** menu command to display the Make Editable Setup form.
4. In the Make Editable Setup form:
   a. Choose the **Incremental** option.
   b. Click **OK**.



Result: The hierarchical design partition views will be created, as seen in the Library Manager. You can now start editing the hierarchical subcell.

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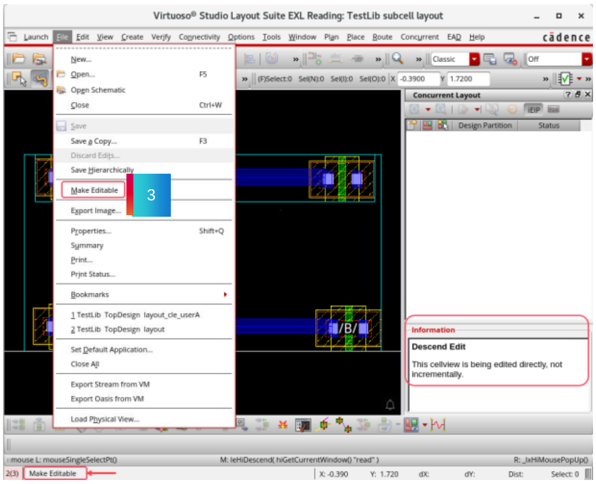
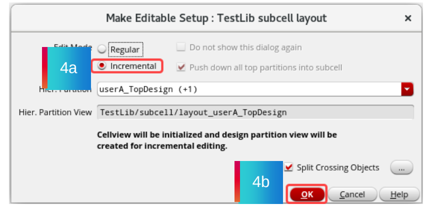
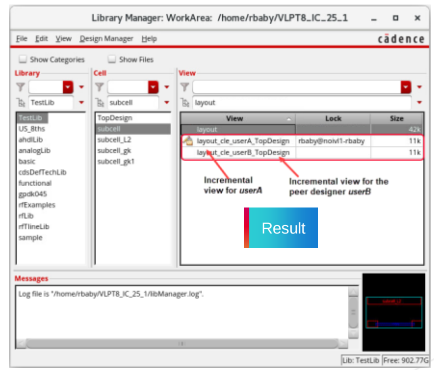
Result: Editing a Subcell Incrementally After Descending in Read Mode







You can choose *Regular* or *Incremental* to start editing the hierarchical subcell.

This page does not contain notes.

Submitting the Incremental EIP Updates for Merge: *TopDesign/layout_cle_userA* and *subcell/layout_cle_userA_TopDesign*

After you have completed and reviewed all the edits in the top design view in the design partition **userA**, you can submit the hierarchical design partition for merge.

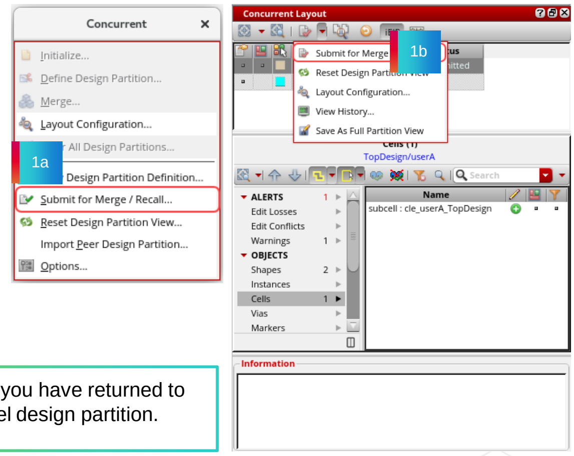
To submit a hierarchical design partition for merge:

1. To open the Submit for Merge form, use any one of the following methods:
   a. In the layout canvas, choose the **Concurrent – Submit for Merge / Recall** menu command.
   b. Click the **Submit for Merge** option on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant to submit the updates made in the subcell.

The top-level design partition *TopDesign/userA* is submitted for merge, and *Submit for Merge* dialog box is displayed.

2. Then, click **OK** in the *Submit for Merge* form.

Result: The status of the top design partition view *TestLib/TopDesign/layout_cle_userA* changes to *Submitted*.



Make sure you have returned to the top-level design partition.

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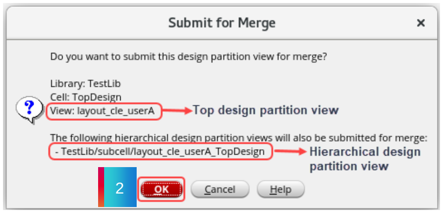
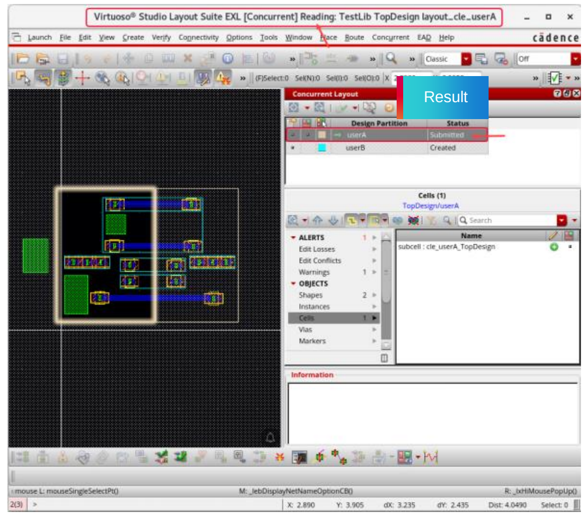
Result: Submitting the Incremental EIP Updates for Merge: *TopDesign/layout_cle_userA* and *subcell/layout_cle_userA_TopDesign*





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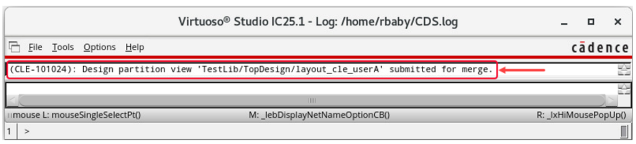
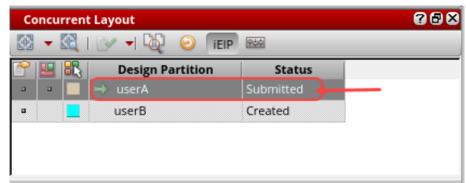
Result: Submitting the Incremental EIP Updates for Merge: *TopDesign/layout_cle_userA* and *subcell/layout_cle_userA_TopDesign* (continued)

Observe the CIW window and *CDS.log* file for the top design partition view *TestLib/TopDesign/layout_cle_userA* merging details.

The *Status* changes to *Submitted* for the top design partition view and all the hierarchical partition views that were edited and were in the *Not Submitted* state.



\o (CLE-101024): Design partition view 'TestLib/TopDesign/layout_cle_userA' submitted for merge.



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Generating the Temporary Pins in Designer Mode

You need to generate temporary pins in the designer mode for space-based routing between design partitions.

These pins are generated only on the nets crossing design partitions.

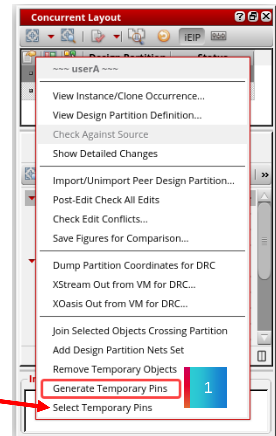
To generate temporary pins:

1. In the layout canvas, **right**-click in the *Concurrent Layout* assistant and choose **Generate Temporary Pins**.

Result: The temporary pins are added to the current design partitions.

To select these temporary pins for deleting or cross-selecting their nets:

Right-click in the Concurrent Layout assistant and choose **Select Temporary Pins**.



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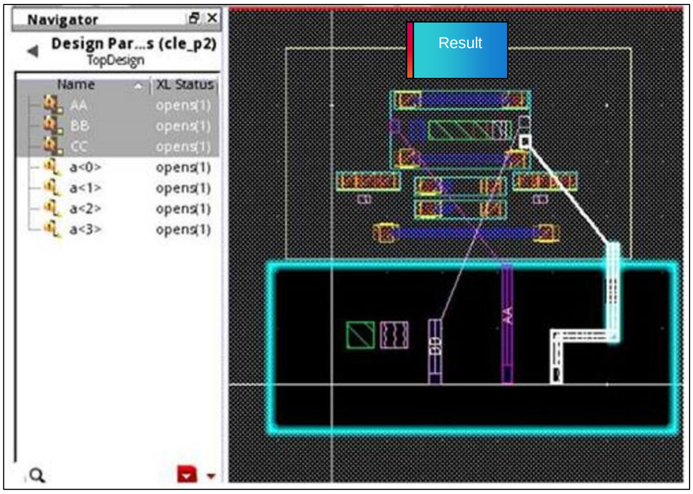
Result: Generating the Temporary Pins in Designer Mode



In this graphic:

- Net **AA** is exactly on the area boundary of the design partition.
- Net **BB** is inside the area boundary.
- Net **CC** is outside the area boundary.
- When you run the *Generate Temporary Pins* command, pins are generated only on the nets **AA** and **CC**.

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Performing Automatic Routing in an Area-Based Design Partition

In an area-based design partition, the scope of edit is defined in terms of area. The behavior of automatic routing varies based on which edit mode is enabled in Layout XL/EXL/MXL tiers. You can use different settings to achieve the intended results.

This example considers two area-based design partitions, **userA** and **userB**, added to a design. To perform automatic routing:

1. Open the area-based design partition **userA** from the Library Manager in Layout EXL.
2. Set edit mode to *Only Edit Inside Partition* in the *Concurrent Layout* assistant.
3. Select all the nets in the Navigator,
   a. **Right**-click
   b. Choose **Route With Default Lookup** from the **Net** context-sensitive menu for Virtuoso® Space-based Routing.

Result: The Virtuoso Space-based Router routes only the pins that are inside the design partition **userA**.

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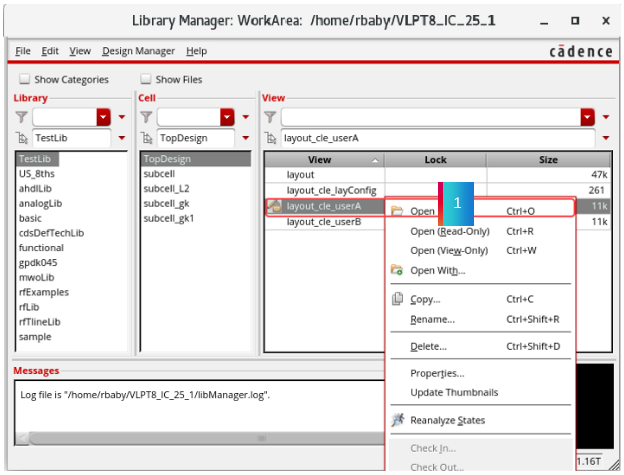
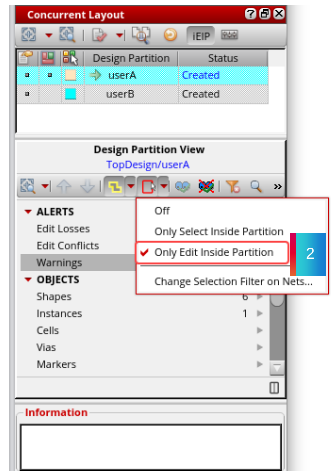
Result: Performing Automatic Routing in an Area-Based Design Partition

The design partition **userA** is opened.

Set **Edit Scope** to *Only Edit Inside Partition*.





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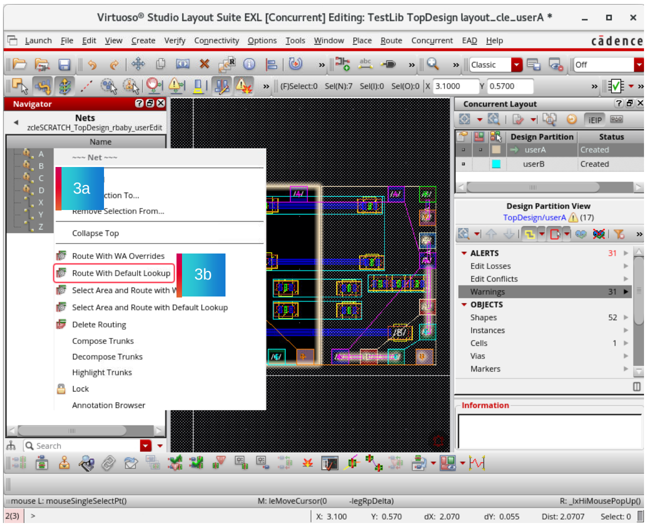
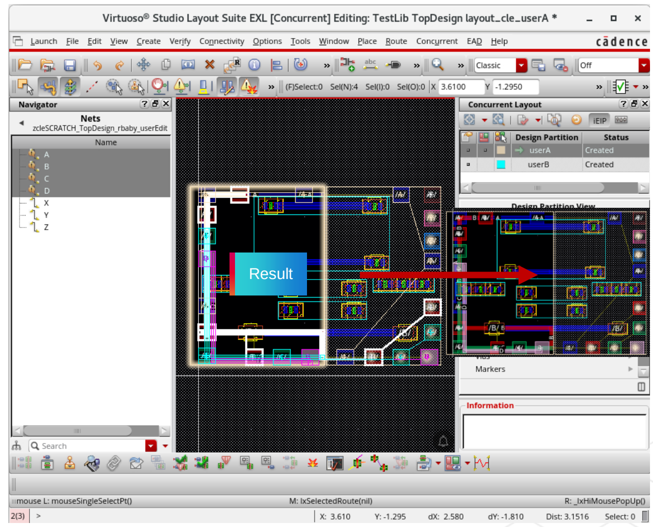
Result: Performing Automatic Routing in an Area-Based Design Partition (continued)

Select all the nets and *Route With Default Lookup*.

Virtuoso Space-based Router routes the nets.





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Automatic Routing in an Area-Based Design Partition: Edit Scope Settings

When you perform routing inside an area-based design partition, you can route:

- Fully between the pins that are inside the design partition.
- Partially, if one pin is inside and the other pin is outside the design partition.

Different designers are expected to work in their respective areas.

- By default, Concurrent Layout sets the edit mode to *Only Select Inside Partition*, which is a moderate level of restriction.
  - The level of restriction varies with different edit modes in the following way:

Edit mode	Level of editing allowed	Description
Off	No restrictions	You can create and edit inside and outside the current design partition.
Only Select Inside Partition	Moderate	You can create objects outside the current design partition but cannot select any object outside your design partition.
Only Edit Inside Partition	Restrictive	You can create, select, and edit only inside the current design partition.

Interactive wire editing supports and respects the above edit modes.

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Performing Automatic Routing in a Layer-Based Design Partition

In a layer-based design partition, the scope of edit is defined in terms of a range of layers. The behavior of automatic routing varies based on which edit mode is enabled in Layout XL/EXL/MXL tiers. You can use different settings to achieve the intended results.

This example considers two layer-based design partitions, **userA** and **userB**, added to a design. To perform automatic routing:

1. Open the layer-based design partition **userA** from the Library Manager in Layout EXL.
2. Set edit mode to *Only Select Inside Partition* in the *Concurrent Layout* assistant.
3. Select all nets in the Navigator,
   a. **Right**-click
   b. Choose *Route With Default Lookup* from the **Net** context-sensitive menu for Virtuoso Space-based Routing.

Result: The Virtuoso Space-based Router routes only on layers that are included in the current design partition **userA**.

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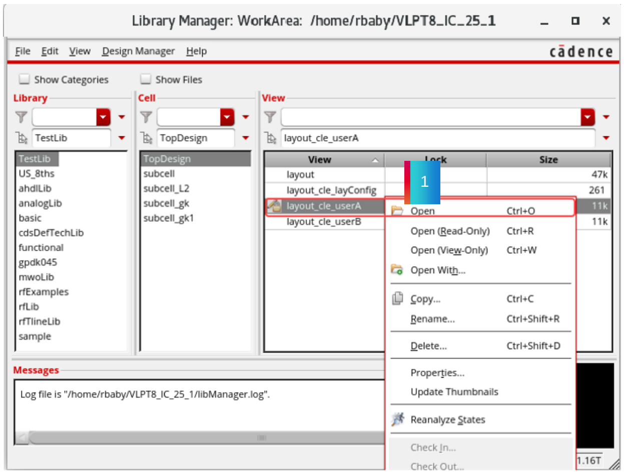
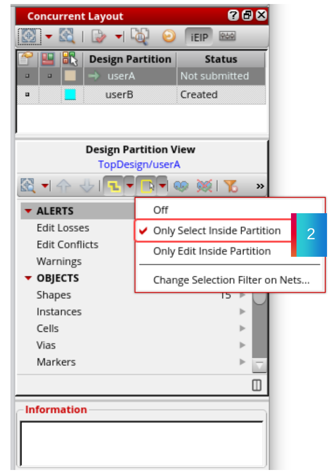
Result: Performing Automatic Routing in a Layer-Based Design Partition

The design partition **userA** is opened.

Set **Edit Scope** to *Only Select Inside Partition*.





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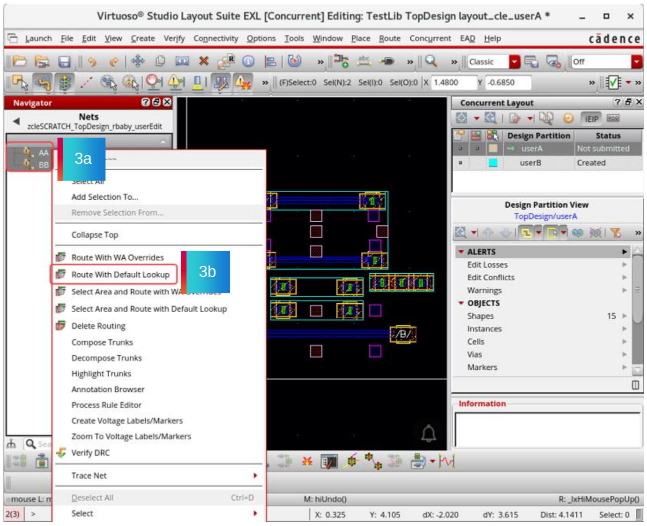
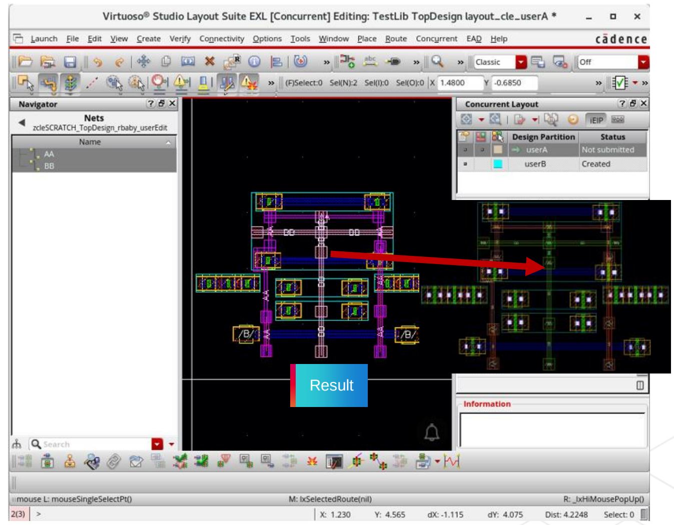
Result: Performing Automatic Routing in a Layer-Based Design Partition (continued)

Select all the nets and *Route With Default Lookup*.

Virtuoso Space-based Router routes the nets.





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Automatic Routing in a Layer-Based Design Partition: Edit Scope Settings

When you perform routing inside a layer-based design partition, you can route:

- Fully between the pins that are between the layers in the design partition.

Different designers are expected to work in their respective sets or range of layers.

- By default, Concurrent Layout offers to keep the edit mode to *Only Select Inside Partition*, which is a moderate level of restriction.
  - The level of restriction varies with different edit modes in the following way:

Edit mode	Level of editing allowed	Description
Off	No restrictions	You can create and edit objects on any layer.
Only Select Inside Partition	Moderate	You can only create and edit objects on layers that are included in the current design partition. However, you can use vias that are outside the range of layers in the current design partition.
Only Edit Inside Partition	Restrictive	You can only create and edit objects on layers, including the vias that are inside the current design partition.

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Module Summary

In this module, you learned how to

- Set the Edit Scope for the design partition userA in designer mode
- Edit the design partition userA using the CLE flow
- Edit the design partition userA hierarchically using the CLE flow
- Verify the incremental eip updates
  - subcell/cle_userA_TopDesign
- Submit the incremental EIP updates for merge
  - TopDesign/layout_cle_userA and subcell/layout_cle_userA_TopDesign
- Perform automatic routing in
  - An area-based design partition
  - A layer-based design partition

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Labs

Lab 4-1 Setting the Edit Scope for the Design Partition userA in Designer Mode

Lab 4-2 Editing the Design Partition userA Using the CLE Flow

Lab 4-3 Editing the Design Partition userA Hierarchically Using the CLE Flow

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Quiz: Options Available in the Edit Scope Section

In the Edit Scope section, what are the options available in the Concurrent Layout assistant?

Select all that apply.

A. Off

B. Only Select Inside Partition

C. Only Edit Inside Partition

D. On

E. All of the above

Answer: A, B, and C

In the Edit Scope section, Off, Only Select Inside Partition, and Only Edit Inside Partition options are available in the Concurrent Layout assistant.

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Quiz: Concurrent Layout Modes

What modes can you use for the Concurrent Layout editing?

Select all that apply.

A. Top-down
B. Bottom-up
C. Single User
D. Multiple User
E. All of the above

Answer: C and D

The Concurrent Layout can be used both in the single-user and multiple-user modes.

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Quiz: Editing in an Area-Based Design Partition

In designer mode, by default, in area-based design partitions, the area boundary for the selected design partition is highlighted, and the region outside it is grayed out.

A. True

B. False

Answer: A

Yes. In designer mode, by default, in area-based design partitions, the area boundary for the selected design partition is highlighted, and the region outside it is grayed out.

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Quiz: Default Option in the Edit Scope Section

In the Edit Scope section, which option is enabled by default in the Concurrent Layout assistant?

Select the correct choice.

A. Off

B. Only Select Inside Partition

C. Only Edit Inside Partition

D. On

E. None of the above

Answer: B

In the Edit Scope section, the Only Select Inside Partition option is enabled by default in the Concurrent Layout assistant.

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Quiz: Editing in a Layer-Based Design Partition

When editing in a layer-based design partition, the Edit Scope is set to the range of layers included in that design partition.

A. True

B. False

Answer: A

Yes. When editing in a layer-based design partition, the Edit Scope is set to the range of layers included in that design partition.

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Module 5

Editing the Design Partition userB in Designer Mode

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Module Objectives

In this module, you

- Import/Unimport the peer design partition updates after the Concurrent Layout editing
- Set the Edit Scope for the design partition userB in designer mode
- Edit the design partition userB using the CLE flow
- Edit the design partition userB hierarchically using the CLE flow
- List out the changes in the hierarchical design partition view
  - subcell/cle_userB_TopDesign
- Verify the incremental EIP updates
  - subcell/cle_userB_TopDesign
- Identify the changes made by the peer users
- Submit the incremental EIP updates for merge
  - TopDesign/layout_cle_userB and subcell/layout_cle_userB_TopDesign

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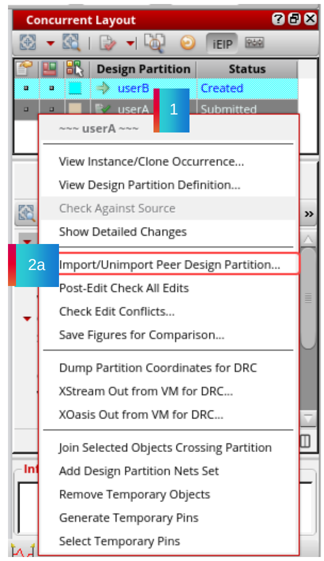
Importing the Peer Design Partition Updates After the Concurrent Layout Editing

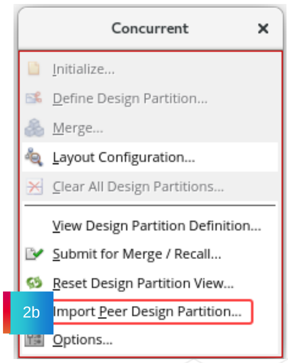
You can use the **Import Peer Design Partition** form to import the changes made in the peer design partitions and view the updated design.

To import the changes made in the peer design partitions and view the updated design as a whole:

1. In the layout canvas, **right**-click the design partition.
2. To open the Import Peer Design Partition form, use any one of the following methods:
   a. Click the **Import/Unimport Peer Design Partition** option on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant.
   b. Choose the **Concurrent – Import Peer Design Partition** menu command.

The Import Peer Design Partition form appears.





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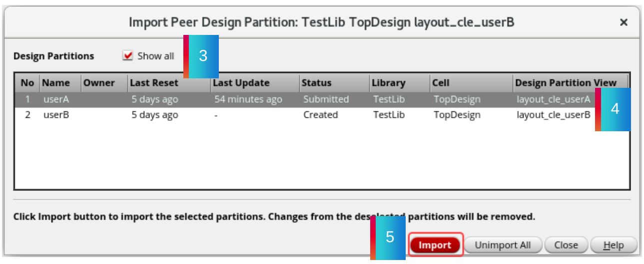
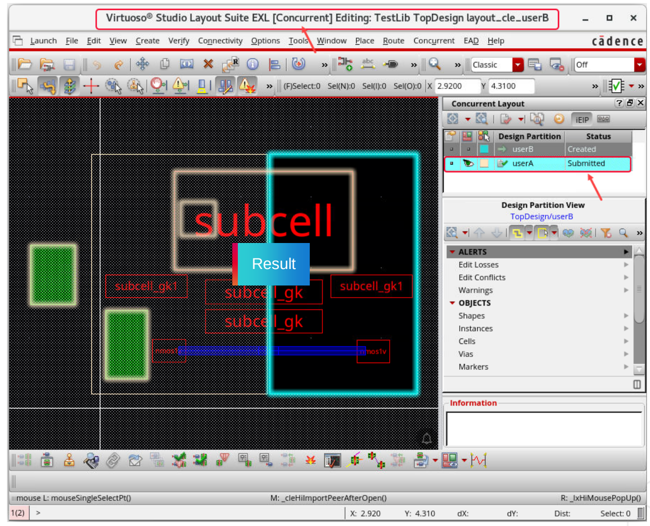
Result: Importing the Peer Design Partition Updates After Concurrent Layout Editing

3. Select **Show all** to display all the edited peer design partitions.
4. Select the design partitions to import.
5. Click the **Import** button.

Result: Updates from the selected design partitions will be imported for your reference.





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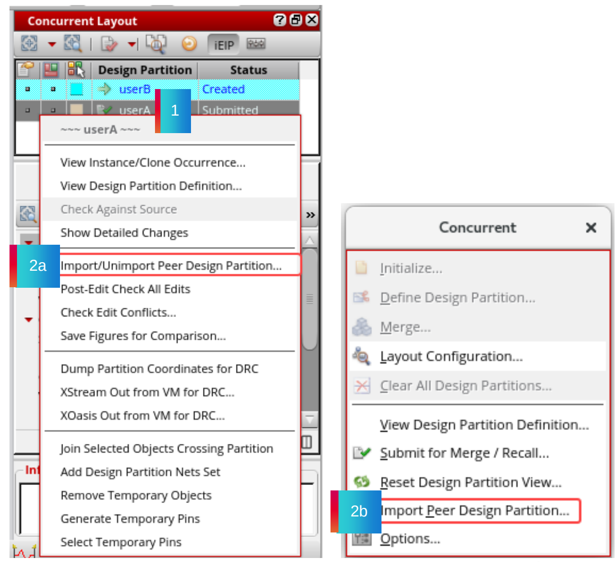
Unimporting the Peer Design Partition Updates

You can remove the imported information related to the updates made in the peer design partitions from the current design partition.

To remove the imported information:

1. In the layout canvas, **right**-click the design partition.
2. To open the Import Peer Design Partition form, use any one of the following methods:
   a. Click the **Import/Unimport Peer Design Partition** option on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant.
   b. Choose the **Concurrent – Import Peer Design Partition** menu command.

The Import Peer Design Partition form appears.



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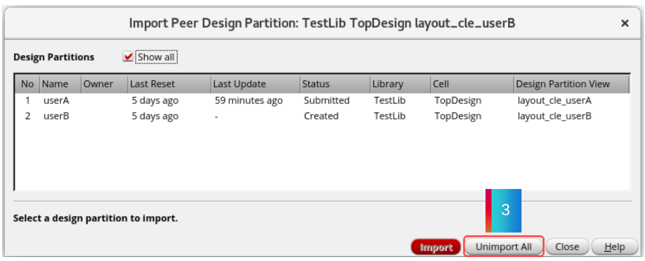
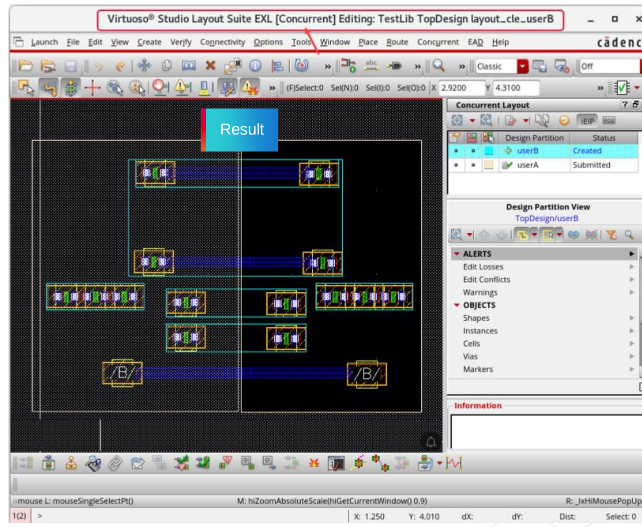
Result: Unimporting the Peer Design Partition Updates

3. Click **Unimport All** in the Import Peer Design Partition form.

Result: All imported updates will be removed from the current design partition.





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Setting the Edit Scope for the Design Partition userB in Designer Mode

The Edit Scope for each design partition is set to select only those objects that are inside the current partition by default.

You can change these settings using one of the following two methods in designer mode:

1. Setting the Edit Scope for **userB** using the *Concurrent Layout Options* Form.
2. Setting the Edit Scope for **userB** using the *Concurrent Layout* Assistant.

Off	Disables the edit mode settings.
Only Select Inside Partition	Lets you select only those objects that are inside the current partition. Objects that are fully or partially inside or have been temporarily moved outside the current partition for editing are also selectable.
Only Edit Inside Partition	Lets you edit only those objects that are inside the current partition. When you specify this option, all edit commands are restricted to the specified design partition. You cannot edit objects outside this design partition.

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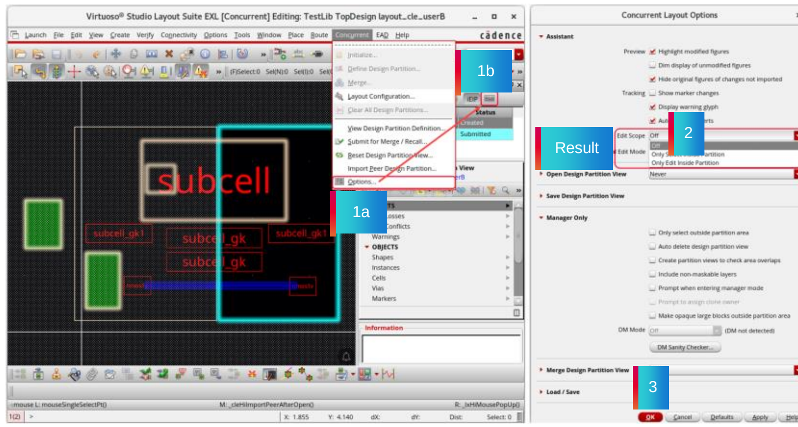
Method 1: Setting the Edit Scope for userB Using the Concurrent Layout Options Form

To set the edit scope for **userB**, perform the following steps:

1. Open the *Concurrent Layout Options* form using any one of the two methods:
   a. From the layout window, choose the **Concurrent – Options** menu command.
   b. Click the **Options** icon in the *Concurrent Layout* assistant to display the *Concurrent Layout Options* form.
2. In the *Concurrent Layout Options* form, in the *Edit Scope* cyclic field, choose the required option. In this case, say, select the option, *Off*.
3. Click **OK** in the form.

Result: *Edit Scope* is set to *Off*.



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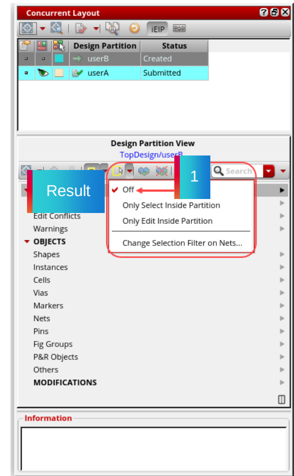
Method 2: Setting the Edit Scope for userB Using the Concurrent Layout Assistant

To set the edit scope for **userB**, do the following steps:

1. In the *Concurrent Layout* assistant, from the *Edit Scope* cyclic field, choose the required option. In this case, say, select the option **Off**.

Result: *Edit Scope* is set to **Off**.



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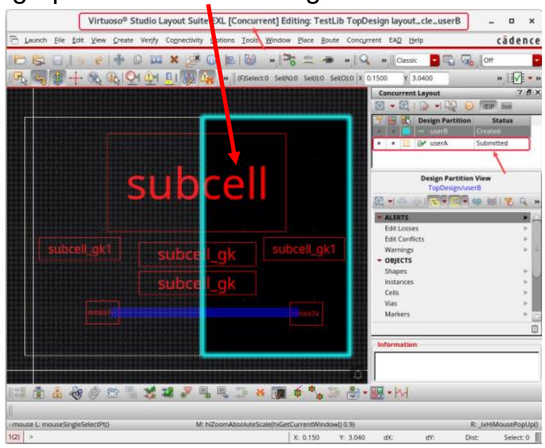
Editing the Design Partition userB Using the CLE Flow

After the required design partition and design partition views are created, designers can start editing their allocated design partition views. Designers should have write permission on the design partition views that store the edits of the top design.

You will explore editing the design partition **userB** using the CLE flow:

1. Editing Inside **userB**
2. Editing Outside **userB**



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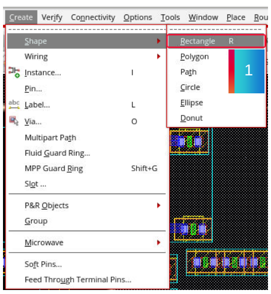
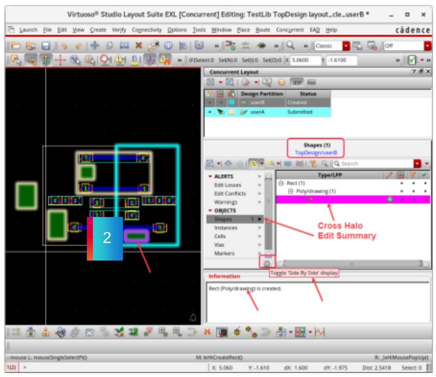
Editing Inside the Design Partition userB Using the CLE Flow

You will make some edits inside the design partition **userB**.

1. As an example, in the layout canvas, choose the **Create – Shape – Rectangle** menu command or press the bindkey **R** to invoke the Create Rectangle command.
2. Create a rectangle inside the design partition, as shown here.





Result

Observe the *Edit Summary* and the *Cross Halo* results for the rectangle shape created inside the design partition **userB** in the *Concurrent Layout* assistant.

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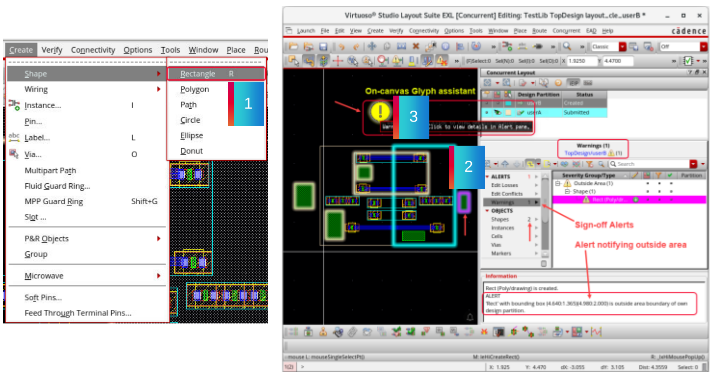
Editing Outside the Design Partition userB Using the CLE Flow

You will make some edits outside the design partition **userB**.

1. As an example, in the layout canvas, choose the **Create – Shape – Rectangle** menu command or press the bindkey **R** to invoke the Create Rectangle command.
2. Create a rectangle outside the design partition, as shown here.
3. Click the *On-canvas Glyph assistant* to view the details in the *Alert* pane.



Result

Observe the *Sign-off Alerts* and the *Alert notifying outside area* result for the rectangle shape created outside the design partition **userB** in the *Concurrent Layout* assistant.

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Removing the Outside Area Edits

You can use the **Remove Changes from Design Partition View** form to remove and discard unwanted edits in your design.

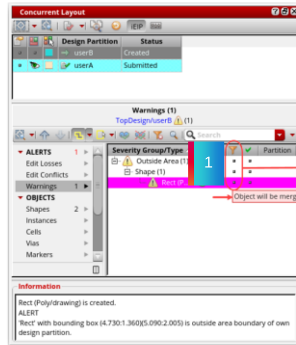
To discard the edits in the design:

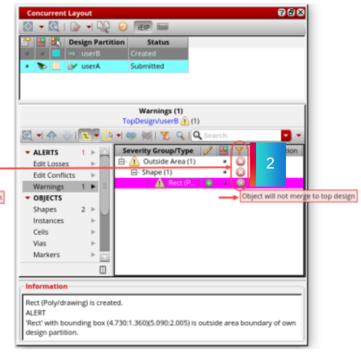
1. In the layout canvas, click the **Filter State** column to mark from *Object will be merged to top design* to *Object will not merge to top design*.
2. Click the *Remove Changes from Design Partition View* button to remove the rectangle added outside the area.

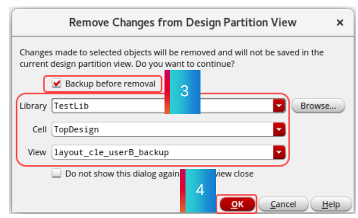
The Removed Changes from Design Partition View form is displayed.

3. Keep the options as required in the Remove Changes from Design Partition View form.
4. Click **OK** in the form.

Result: The rectangle added outside the area boundary is removed from the layout canvas.







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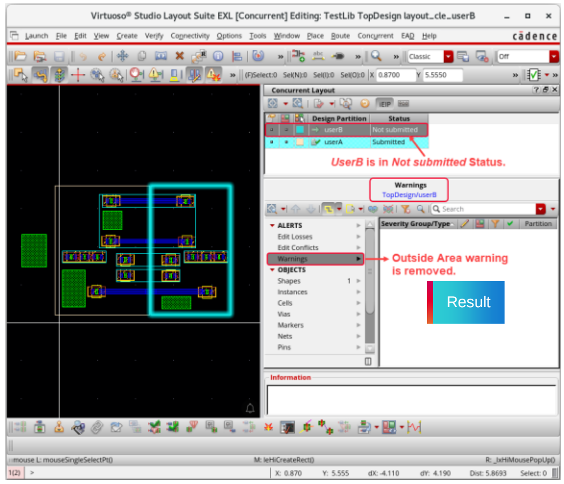
Result: Removing the Outside Area Edits



The *Alert notifying outside area* results for the rectangle shape created outside the design partition **userB** in the *Concurrent Layout* assistant is removed.

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Editing the Design Partition userB Hierarchically Using the CLE Flow

Hierarchical editing in the CLE flow lets multiple designers concurrently Edit In Place (EIP) into hierarchical subcells.

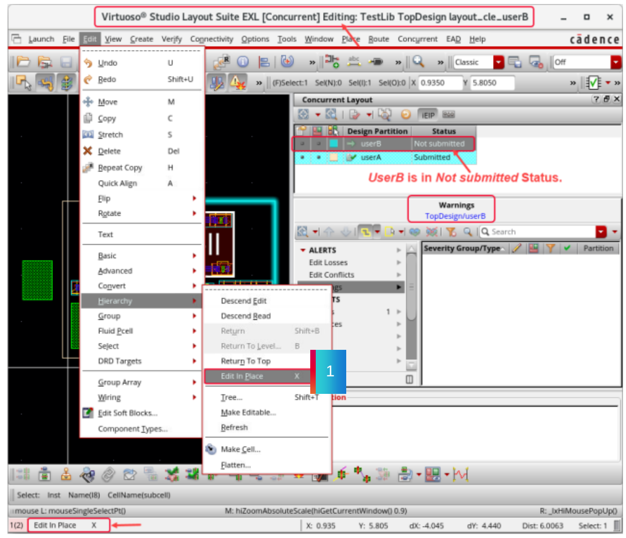
You will make hierarchical editing of the design partition **userB** using the CLE flow. You will descend into the instance *subcell* and do hierarchical editing on it.

1. In the layout canvas,
   - Select the instance *subcell* and choose the **Edit – Hierarchy – Edit In Place** menu command.

   or

   - Press the bindkey **X** to EIP to descend into the selected instance *subcell*.
2. Create a shape in the hierarchical design partition view *layout_cle_userB_TopDesign*.

Result: You have edited the hierarchical instance *subcell* using the CLE flow.



You are descending into *TestLib/subcell/layout_cle_userB_TopDesign* cellview.

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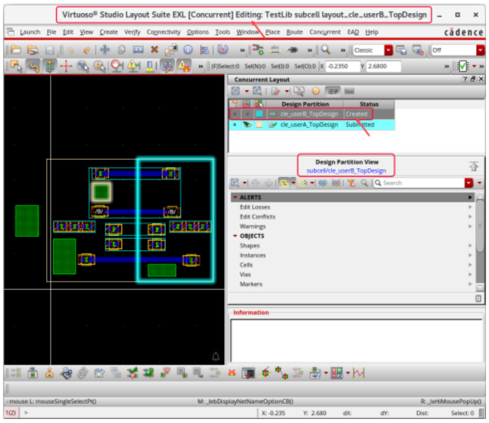
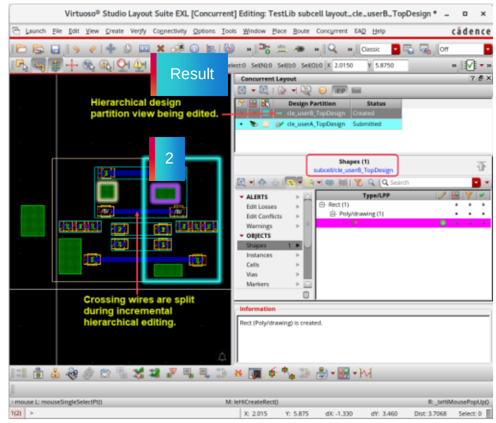
Result: Editing the Design Partition userB Hierarchically Using the CLE Flow

You are now descended into *TestLib/subcell/layout_cle_userB_TopDesign* cellview.



You have edited the hierarchical instance *subcell* using the CLE flow.



The user who first EIPs into the hierarchical subcell in incremental mode creates hierarchical design partition views for themselves as well as for the peer designers. Therefore, the *userB* directly opens the hierarchical design partition view that was created by *userA* in incremental mode. Thus, the *Hierarchical Edit Setup* form is NOT displayed again.

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Returning to the Top Partition Cellview: TestLib/TopDesign/layout_cle_userB

After the hierarchical editing (EIP) in the CLE flow is completed, designers will return to the top partition cellview.

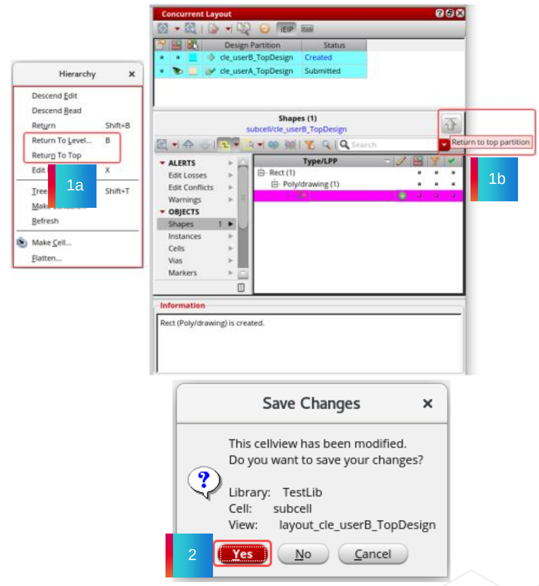
To return to the top partition cellview:

1. First, select any one of the two methods:
   a. In the layout canvas, choose the **Edit – Hierarchy – Return To Level** (bindkey **B**) / **Return To Top** menu command.
   b. Click the **Return to top partition** button in the *Concurrent Layout* assistant.

You get the *Save Changes* form to inform you to save the changes you have made in the hierarchical design partition view *layout_cle_userB_TopDesign*.

2. In the *Save Changes* form, click the **Yes** button to save the changes in the hierarchical design partition view *layout_cle_userB_TopDesign*.

Result: You are now returned to the top design partition view *TopDesign/userB*.



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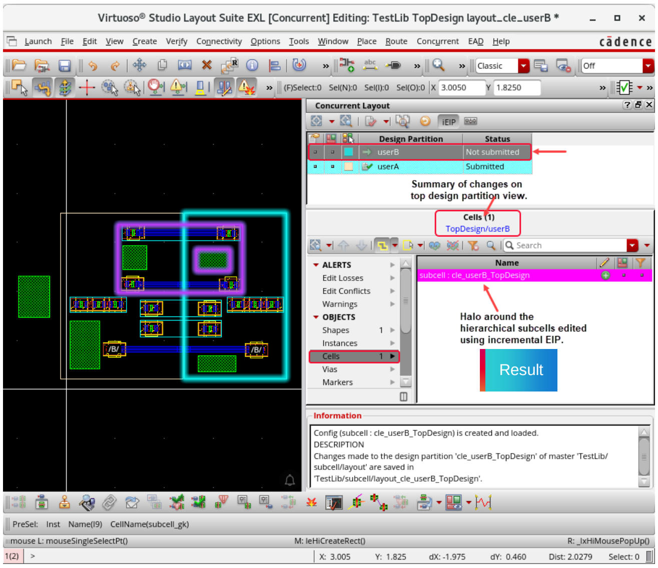
Result: Returning to the Top Partition Cellview:
TestLib/TopDesign/layout_cle_userB



CLE assistant summarizes all the edits performed in the top design partition view (in this case, *TopDesign/userB*) and incrementally edited hierarchical subcells (in this case, *subcell/cle_userB_TopDesign)*. The hierarchical subcells edited in regular mode are excluded.

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Summarizing the Changes in the Hierarchical Design Partition View: subcell/cle_userB_TopDesign

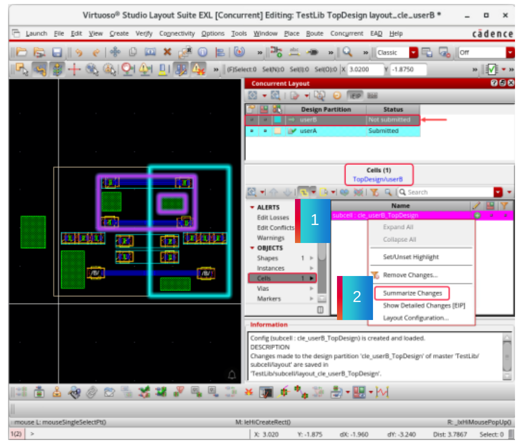
You can summarize the changes inside the incrementally edited hierarchical subcell.

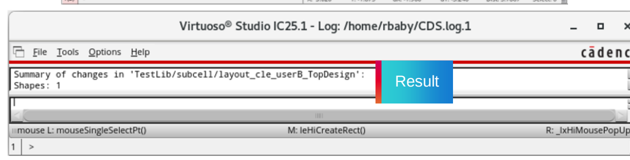
To observe the changes done in the hierarchical design partition view:

1. In the layout canvas, **right**-click the object representing the hierarchical design partition view *subcell:cle_userB_TopDesign*.
2. Choose the **Summarize Changes** option.

Result: The summary of changes is listed in the CIW window and *CDS.log* file.

Observe the CIW window and *CDS.log* file for the *Summary of changes* details in the hierarchical design partition view *TestLib/subcell/layout_cle_userB_TopDesign*.






```
\a _cleHiAssistantSummarizeChanges("TestLib" "subcell" "layout_cle_userB_TopDesign")
\o Summary of changes in 'TestLib/subcell/layout_cle_userB_TopDesign':
\o Shapes: 1
```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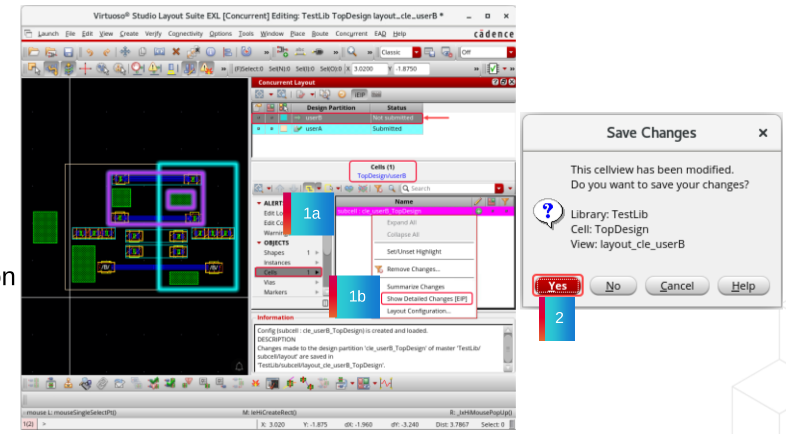
Verifying the Incremental EIP Updates in subcell/cle_userB_TopDesign

After you have completed the hierarchical editing, you can view and verify the incremental Edit In Place updates in the hierarchical instance subcell in the design partition **userB** using the **Show Detailed Changes (EIP)** option.

1. To view and verify the details of the edit changes inside the incrementally edited hierarchical subcell:
   a. In the layout canvas, **right**-click the object representing the hierarchical design partition view *subcell:cle_userB_TopDesign*.
   b. Choose the **Show Detailed Changes (EIP)** option.
2. Save the changes in the top design partition view *TopDesign/userB* by clicking the **Yes** button.



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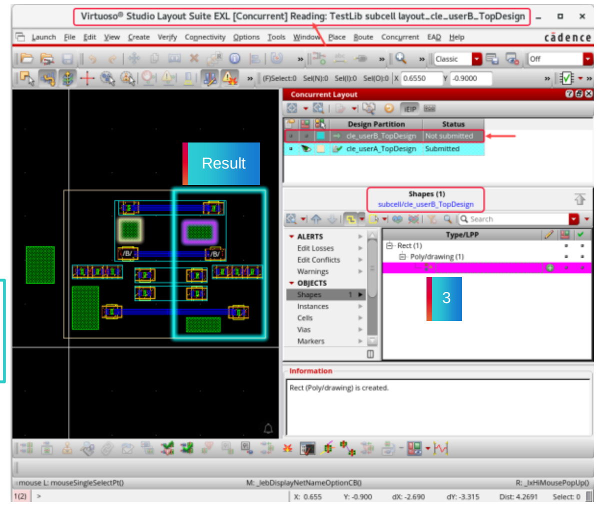
Result: Verifying the Incremental EIP Updates in subcell/cle_userB_TopDesign

3. On the *Concurrent Layout* assistant, expand the correct category to view the edit changes saved in the hierarchical design partition view *subcell/cle_userB_TopDesign*.

Result: You have viewed and verified the incremental EIP updates in *subcell/cle_userB_TopDesign*.



EIP will be performed in read-only mode. The object is shown in the *Cells* category in the *OBJECTS* section of the *Summary* Pane.

When you choose **Show Detailed Changes [EIP]**, EIP is done into the hierarchical design partition view *subcell/cle_userB_TopDesign* in read-mode.

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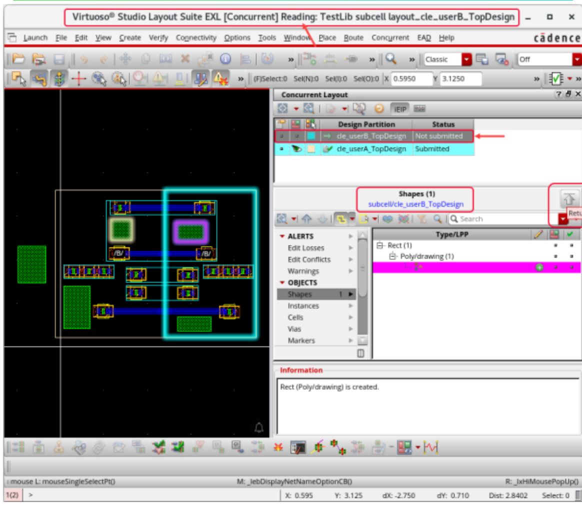
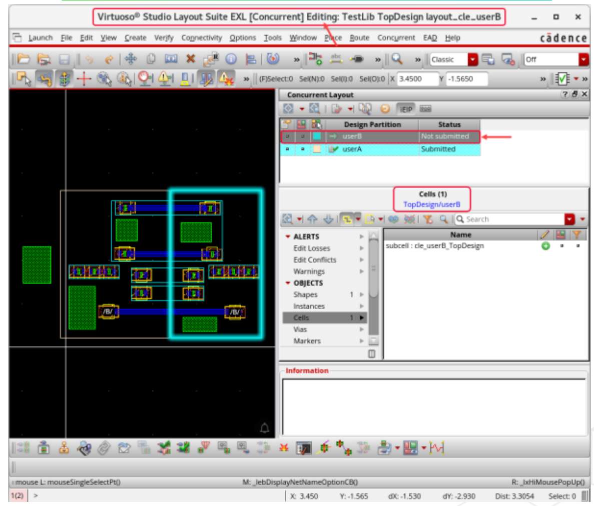
Returning to the Top Design Partition View TopDesign/userB

After reviewing all the changes, click the **Return to top partition** button on the *Concurrent Layout* assistant to return to the top design partition view.



Now you are returned to the top design partition view *TopDesign/userB*.



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Previewing the Changes Made by the Peer Users

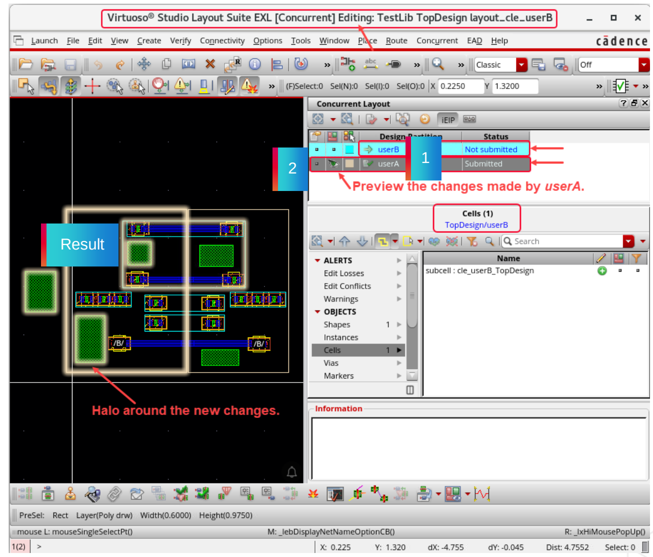
You can preview the changes from your peer designers to check their progress.

To preview the changes in the design partition, **userA**:

1. In the layout canvas, on the *Concurrent Layout* assistant, select the design partition **userA**.
2. Click the **Preview** button.

Result: All the changes made by the design partition **userA** are marked by *Halo* in the canvas. These changes include the changes done in both the top design partition view and the hierarchical design partition view.

Before you import the changes, ensure that your peers have saved their latest changes.



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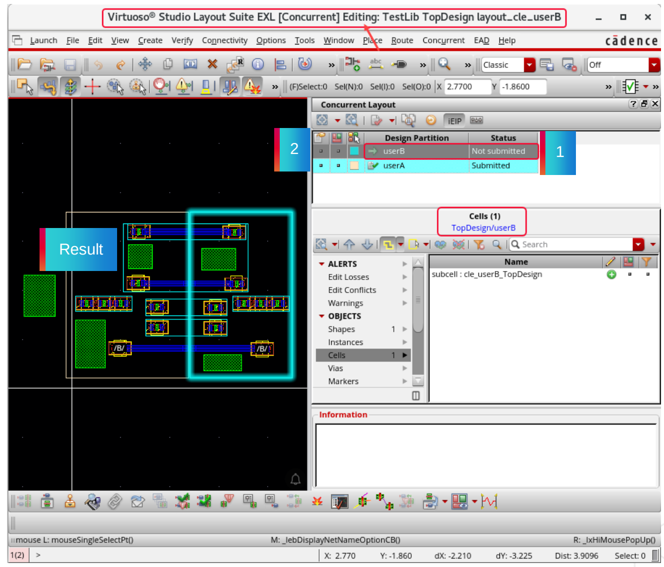
Removing the Preview Selections

You can remove the preview selections made earlier.

To remove the preview:

1. In the layout canvas, on the *Concurrent Layout* assistant, select the design partition **userB**.
2. Unselect the **Preview** button to deselect the selections made earlier.

Result: The previews made earlier are deselected.



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Submitting the Incremental EIP Updates for Merge:
TopDesign/layout_cle_userB and subcell/layout_cle_userB_TopDesign

After you have completed and reviewed all the edits in the top design view in the design partition **userB**, you can submit the hierarchical design partition for merge.

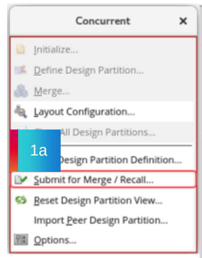
To submit a hierarchical design partition for merge:

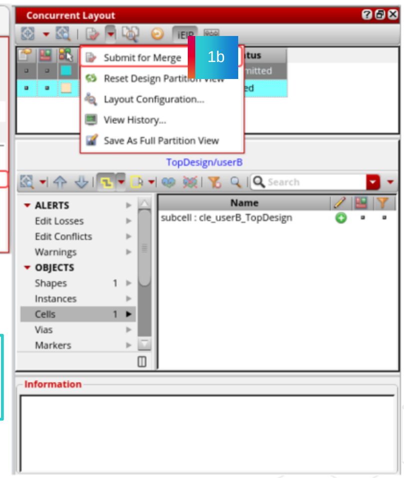
1. To open the Submit for Merge form, use any one of the following methods:
   a. In the layout canvas, choose the **Concurrent – Submit for Merge / Recall** menu command.
   b. Click the **Submit for Merge** option on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant to submit the updates made in the subcell.

The top-level design partition *TopDesign/userB* is submitted for merge, and *Submit for Merge* dialog box is displayed.

2. Click **OK** in the *Submit for Merge* form.

Result: The status of the top design partition view *TestLib/TopDesign/layout_cle_userB* changes to *Submitted*.





Make sure you have returned to the top-level design partition.

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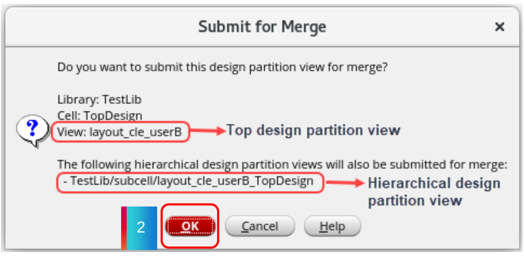
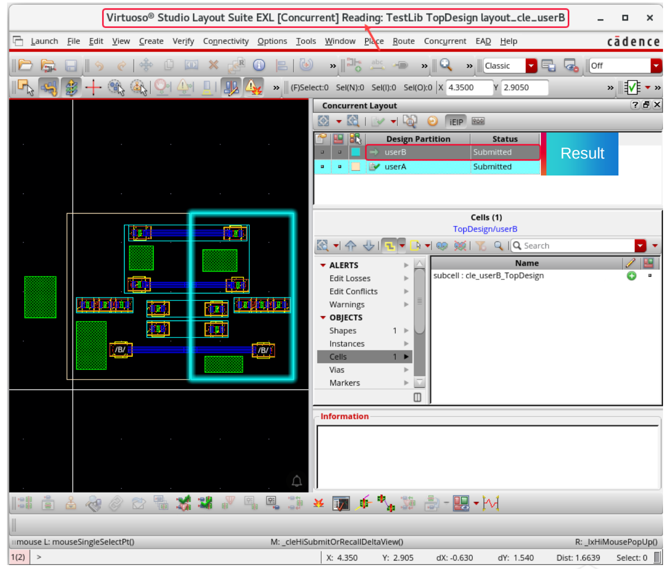
Result: Submitting the Incremental EIP Updates for Merge:
TopDesign/layout_cle_userB and subcell/layout_cle_userB_TopDesign





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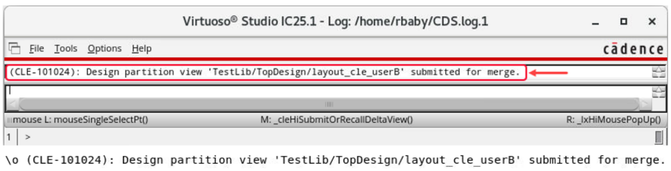
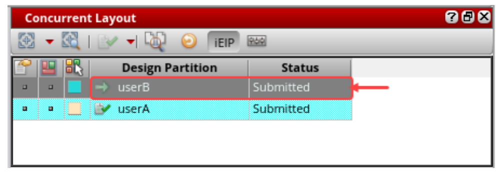
Result: Submitting the Incremental EIP Updates for Merge:
TopDesign/layout_cle_userB and subcell/layout_cle_userB_TopDesign (continued)

Observe the CIW window and CDS.log file for the top design partition view *TestLib/TopDesign/layout_cle_userB* merging details.

The status changes to *Submitted* for the top design partition view and all the hierarchical partition views that were edited and were in the *Not Submitted* state.





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Recalling a Design Partition

You can recall a design partition that is submitted for merge.

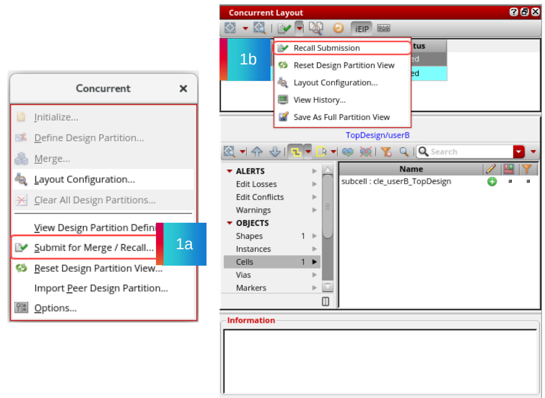
To recall a design partition submitted for merge:

1. To open the Recall Submission form, use any one of the following methods:
   a. In the layout canvas, choose the **Concurrent – Submit for Merge / Recall** menu command.
   b. Click the **Recall Submission** button on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant to recall the design partition.

The *Recall Submission* dialog box opens.

2. Click **OK** in the *Recall Submission* form.

Result: The design partition will be recalled, and the status will change to *Not Submitted*.



You can recall a design partition only if the manager has not yet merged it.

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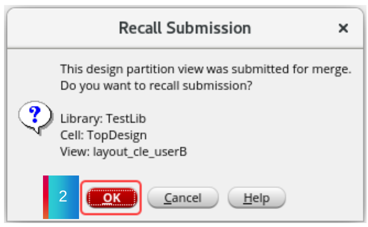
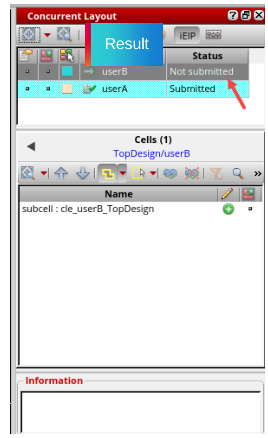
Result: Recalling a Design Partition

The design partition will be recalled, and the status will change to *Not Submitted*.





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Checking and Resolving Edit Conflicts After Concurrent Layout Editing

Use the **Check Edit Conflicts** command to identify even those edit conflicts that were undetected by the assistant alert system, including hierarchical edits. Use this command to thoroughly check for conflicts between the selected design partitions and the top design.

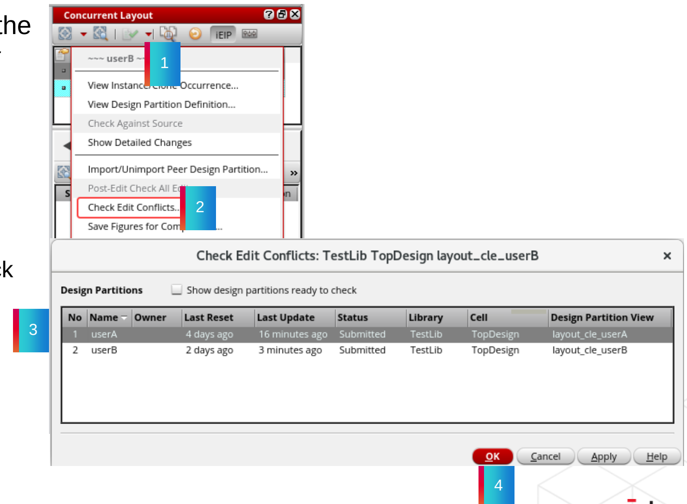
To check the log of all conflicts, follow the steps on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant in the layout canvas.

1. **Right**-click on the design partition.
2. Click the **Check Edit Conflicts** option.

The *Check Edit Conflicts* form is displayed.

3. Select the design partition for which you want to check the edit conflicts.
4. Click **OK** in the *Check Edit Conflicts* form.

Result: The design is checked for edit conflicts, the *Edit Conflicts Summary* report is displayed, and the *Check Edit Conflicts* form is closed.



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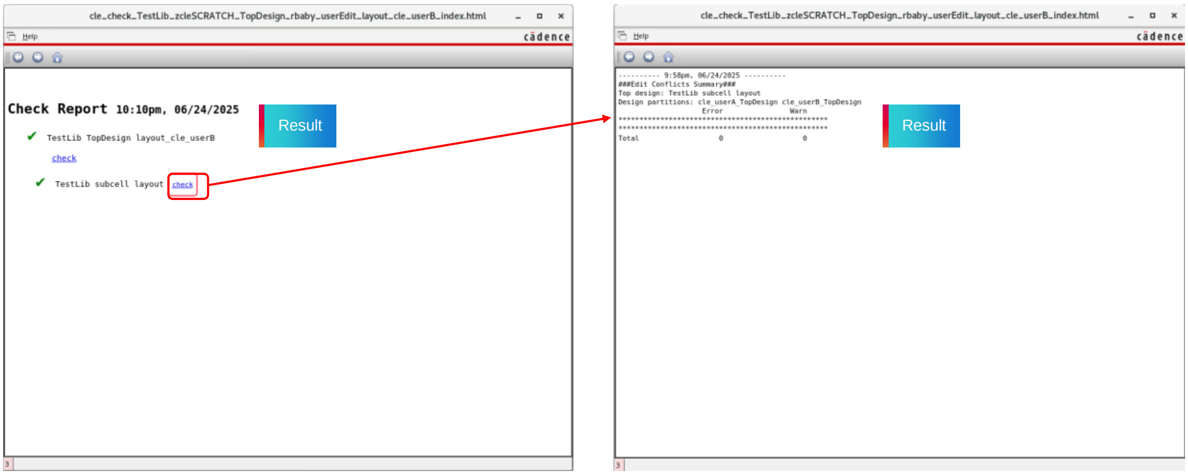
Result: Checking and Resolving Edit Conflicts After Concurrent Layout Editing



You should either resolve or sign off edit conflicts before submitting your design for merge. You can view alerts and warnings and then either resolve or sign off these conflicts from the *Summary Pane* and *Details Pane* of the *Concurrent Layout* assistant.

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Module Summary

In this module, you learned how to

- Import/Unimport the peer design partition updates after the Concurrent Layout editing
- Set the Edit Scope for the design partition userB in designer mode
- Edit the design partition userB using the CLE flow
- Edit the design partition userB hierarchically using the CLE flow
- List out the changes in the hierarchical design partition view
  - subcell/cle_userB_TopDesign
- Verify the incremental EIP updates
  - subcell/cle_userB_TopDesign
- Identify the changes made by the peer users
- Submit the incremental EIP updates for merge
  - TopDesign/layout_cle_userB and subcell/layout_cle_userB_TopDesign

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Labs

Lab 5-1 Setting the Edit Scope for the Design Partition userB in Designer Mode

Lab 5-2 Editing the Design Partition userB Using the CLE Flow

Lab 5-3 Editing the Design Partition userB Hierarchically Using the CLE Flow

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Quiz: Importing the Design Partitions

You can use the Import ____ Design Partition form to import the changes made in the peer design partitions and view the updated design.

Enter the answer in the blank field.

Answer: Peer

You can use the Import Peer Design Partition form to import the changes made in the peer design partitions and view the updated design.

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Quiz: Removing the Imported Information

You can not remove the imported information related to the updates made in the peer design partitions from the current design partition.

A. True

B. False

Answer: B

Yes. You can remove the imported information related to the updates made in the peer design partitions from the current design partition.

This page does not contain notes.

Quiz: Setting the Edit Scope for the Design Partition in Designer Mode

What methods can you use to set the Edit Scope for the design partition in designer mode?

Select all that apply.

A. Concurrent Layout Options form

B. Concurrent Layout assistant

C. Define Design Partition form

D. Annotation Browser assistant

E. All of the above

Answer: A and B

You can use the Concurrent Layout Options form and the Concurrent Layout assistant to set the Edit Scope for the design partition in designer mode.

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Quiz: Eliminating the Unwanted Edits

Which form can you use to eliminate the unwanted edits in your design?

Select the correct choice.

A. Remove Changes from Design Partition View

B. Discard Changes from Design Partition View

C. Delete Changes from Design Partition View

D. Abolish Changes from Design Partition View

E. None of the above

Answer: A

You can use the Remove Changes from Design Partition View form to remove and discard the unwanted edits in your design.

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Quiz: Returning to the Top Partition Cellview

What options can you use to return to the top partition cellview?

Select all that apply.

A. Return To Level
B. Return To Top
C. Return to top partition
D. Return to current partition
E. All of the above

Answer: A, B, and C

You can use the Return To Level, Return To Top, and Return to top partition options to return to the top partition cellview.

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Module 6

Merging/Committing the Top Design in Manager Mode

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Module Objectives

In this module, you

- List out the design partitions userA and userB in the top design
- Identify all the changes in the design partitions userA and userB in the top design
- Merge the submitted design partitions userA and userB
- Commit the merged design partitions userA and userB
- Clear the design partitions userA and userB in Manager mode

This page does not contain notes.

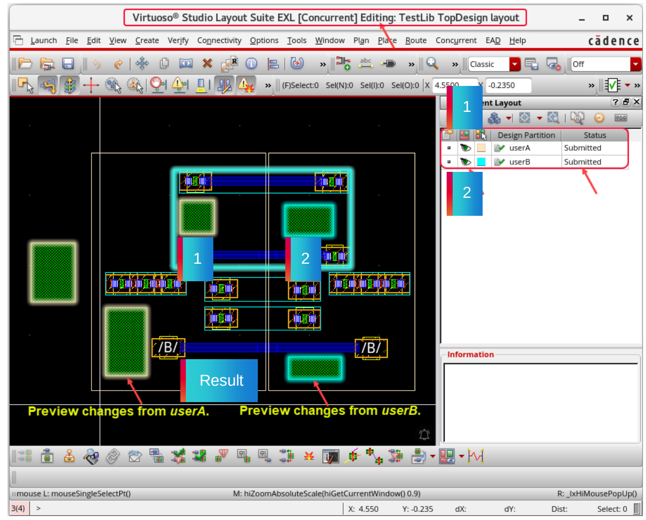
Previewing the Design Partitions userA and userB in the Top Design

You can preview the design partitions from their top design to check their progress.

To preview the design partitions **userA** and **userB** in the top design:

1. In the layout canvas, on the *Concurrent Layout* assistant, select the first design partition **userA** and click the **Preview** button to review the changes.
2. Repeat this step for the second design partition, **userB**.

Result: The new changes appear transiently and are highlighted with a *Halo* in the layout canvas.



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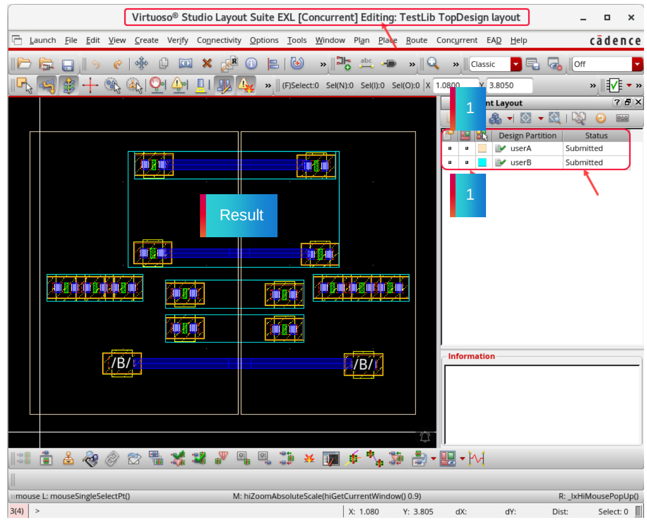
Removing the Previews in the Design Partitions userA and userB in the Top Design

You can remove the preview selections made earlier.



To remove the previews in the design partitions **userA** and **userB** in the top design:

1. In the layout canvas, on the *Concurrent Layout* assistant, unselect the **Preview** button in the design partitions **userA** and **userB** to deselect the selections made earlier.

Result: The previews in the design partitions **userA** and **userB** in the top design are removed.

This page does not contain notes.

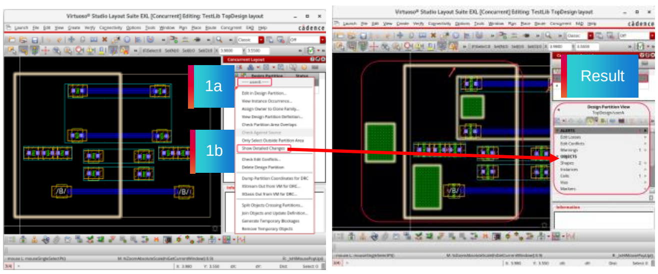
Reviewing All the Changes in the Design Partitions userA and userB in the Top Design

You can review all the changes in detail in the design partitions in the top design using the **Show Detailed Changes** option in the *Concurrent Layout* assistant.

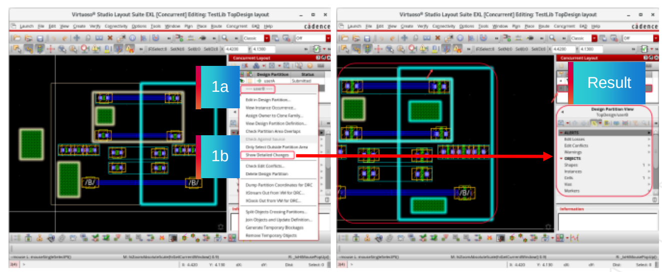
To review all the changes in detail in the design partitions **userA** and **userB** in the top design:

1. In the layout canvas, on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant:
   a. **Right**-click the design partition **userA/userB**.
   b. Select the **Show Detailed Changes** option.

Result: The *Summary of the changes* is loaded in the *Concurrent Layout* assistant for each object type, as shown here.



The *Summary of the changes* in the design partition **userA**.



The *Summary of the changes* in the design partition **userB**.

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Merging the Submitted Design Partitions userA and userB

After the designers have completed editing the design and submitted their respective design partitions for merge, the manager needs to review these requests and then either approve or reject the merge requests.

To approve the merge requests of **userA** and **userB**:

1. To open the Merge form, use any one of the following methods:
   a. In the layout canvas, choose the **Concurrent – Merge** menu command.
   b. Click the **Merge/Un-Merge** option on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant.

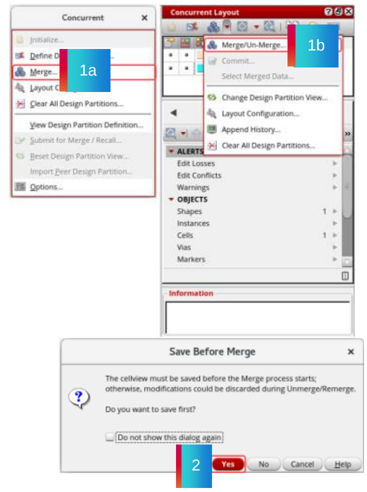
You get the *Save Before Merge* form to save the *TestLib/TopDesign/layout* cellview.

2. Click **Yes** in the *Save Before Merge* form.

Result1: The *TestLib/TopDesign/layout* cellview is saved.

Result2: The *Merge* form is displayed.

You can reject the merge requests using the **Reject Submission** form.



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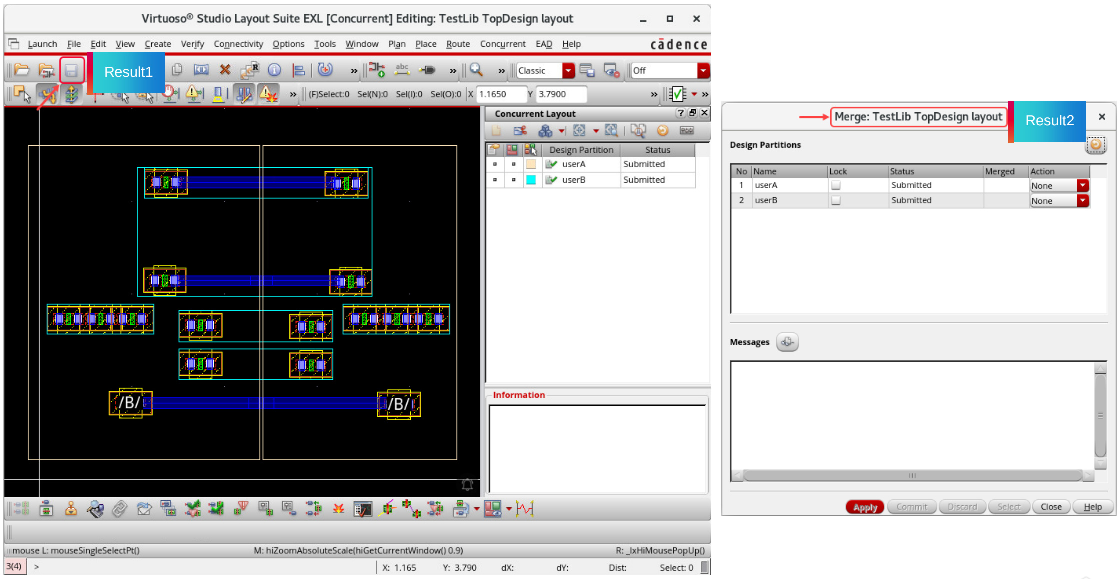
Result: Merging the Submitted Design Partitions userA and userB



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Result: Merging the Submitted Design Partitions userA and userB

(continued)

You will merge the submitted design partitions **userA** and **userB** using the *Merge* form. In the *Merge* form, do the following steps to merge the design partitions:

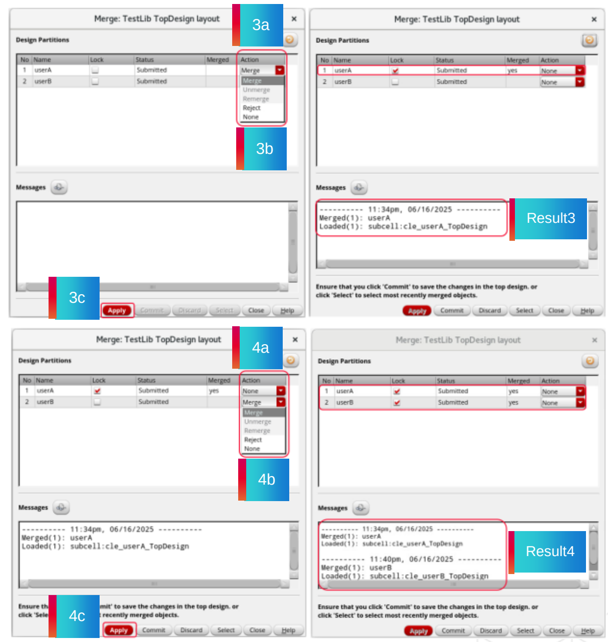
3. To merge the design partition **userA**:
   a. Click in the **Action** field of the design partition **userA**.
   b. Select the **Merge** option.
   c. Click the **Apply** button.

Result3: The design partition view *TestLib/TopDesign/layout_cle_userA* was successfully merged with design partition **userA**.

4. To merge the design partition **userB**:
   a. Click in the **Action** field of the design partition **userB**.
   b. Select the **Merge** option.
   c. Click the **Apply** button.

Result4: The design partition view *TestLib/TopDesign/layout_cle_userB* was successfully merged with design partition **userB**.

Result5: The status of the design partitions **userA** and **userB** changes to *Merged*.



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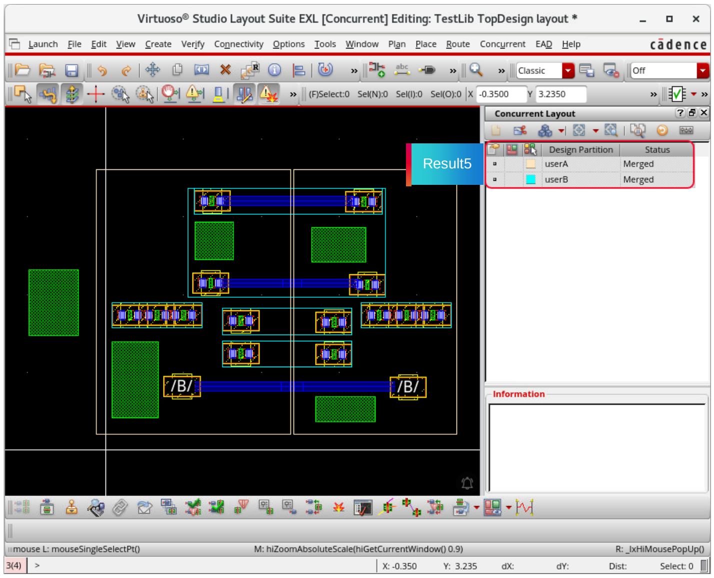
Result: Merging the Submitted Design Partitions userA and userB
(continued)



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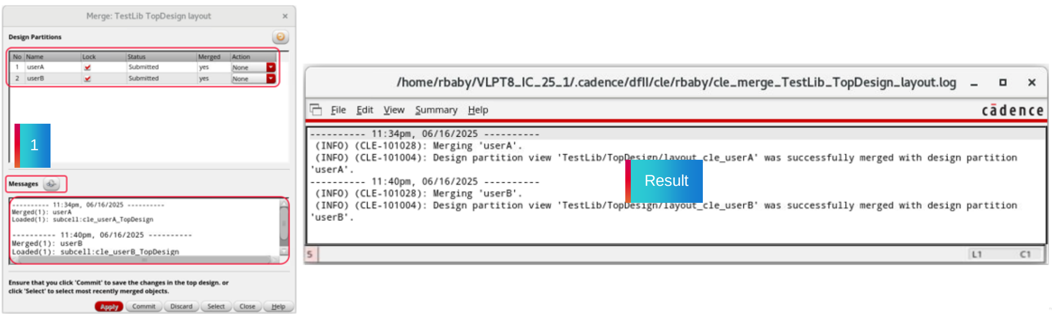
Viewing the Merge Log

Information regarding the actions taken are added to the **Messages** box in the *Merge* form.

To view the merge log:

1. In the *Merge* form, click the **Messages** button to view the merge log.

Result: The merged log is displayed here.



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Committing the Merged Design Partitions userA and userB

You need to run the **Commit** command to complete the merge process and save the merged changes into the top design.

To commit the design partitions:

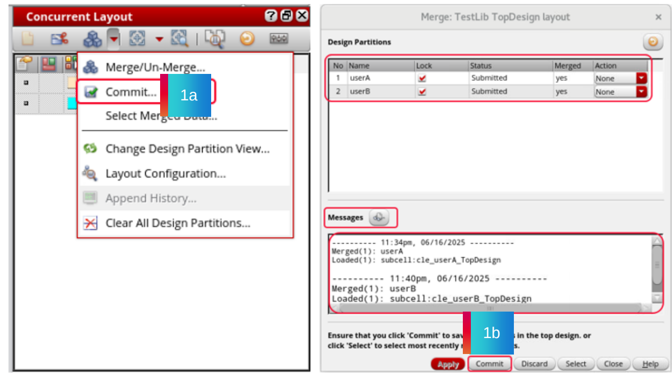
1. To open the Commit form, use any one of the following methods:
   a. In the layout canvas, click the **Commit** option on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant.
   b. Click the **Commit** button in the *Merge* form.

The *Commit* dialog box is displayed to confirm the merged design partitions for **user A and user B** undergoing *Commit*.

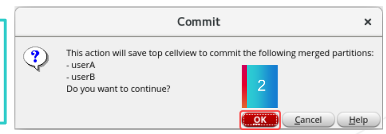
2. Click **OK** in the *Commit* dialog box to continue.

This step recursively merges the children's hierarchical design partition views into the subcell and commits the changes.

Result: The status of the design partitions **userA** and **userB** changes to *Committed*.



The changes from respective design partition views have been added to and saved in the top design.



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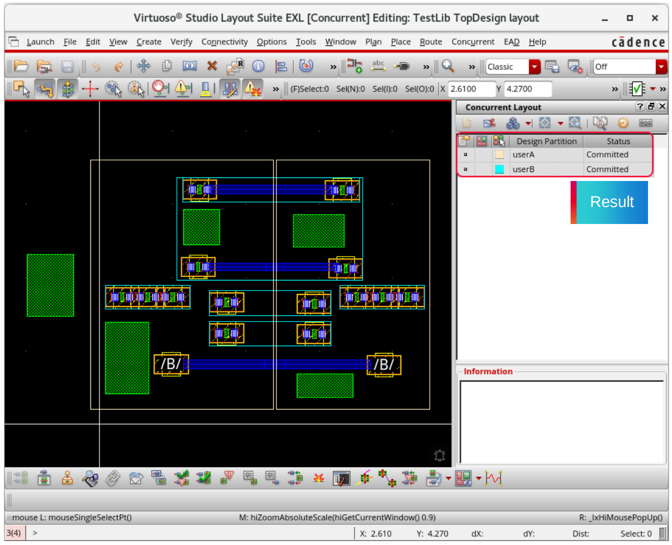
Result: Committing the Merged Design Partitions userA and userB



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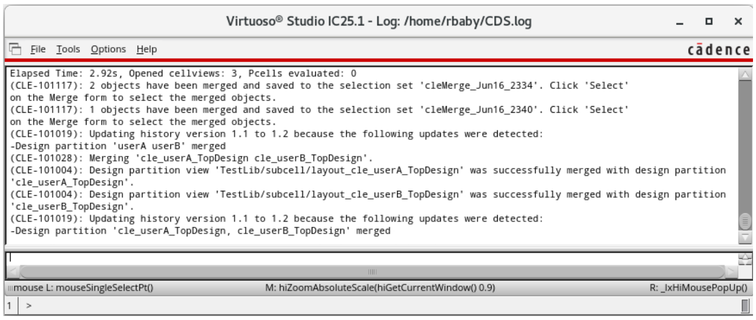
Result: Committing the Merged Design Partitions userA and userB

(continued)

Observe the CIW window for the merging details of the design.



After committing, the design partition views still exist. However, the edit locks on them are released. Designers can continue to work on these design partitions if needed.

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Clearing the Design Partitions userA and userB in Manager Mode

After all the design partitions are merged with the top design, you might want to delete all design partitions and exit the Concurrent Layout Editing environment. If your task is complete and there is no need for concurrent editing, you can delete the design partitions.

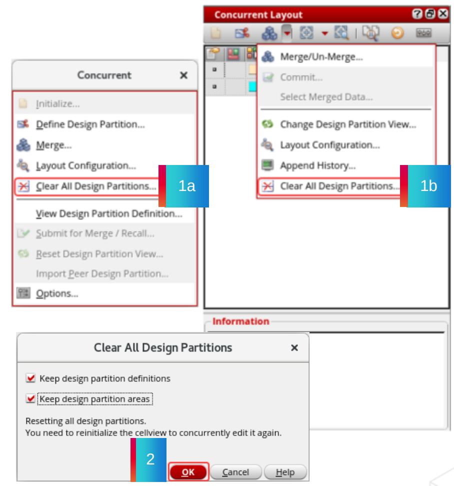
To clear all the design partitions:

1. To open the Clear All Design Partitions form, use any one of the following methods:
   a. In the layout canvas, choose the **Concurrent – Clear All Design Partitions** menu command.
   b. Click the **Clear All Design Partitions** option from the *Merge* dropdown list on the *Concurrent Layout* toolbar in the *Concurrent Layout* assistant to clear all the design partitions.

The Clear All Design Partitions form is displayed.

2. With the required options selected, click **OK** on the form.

Result: All the design partitions will be deleted, and you can edit the design without using Concurrent Layout.



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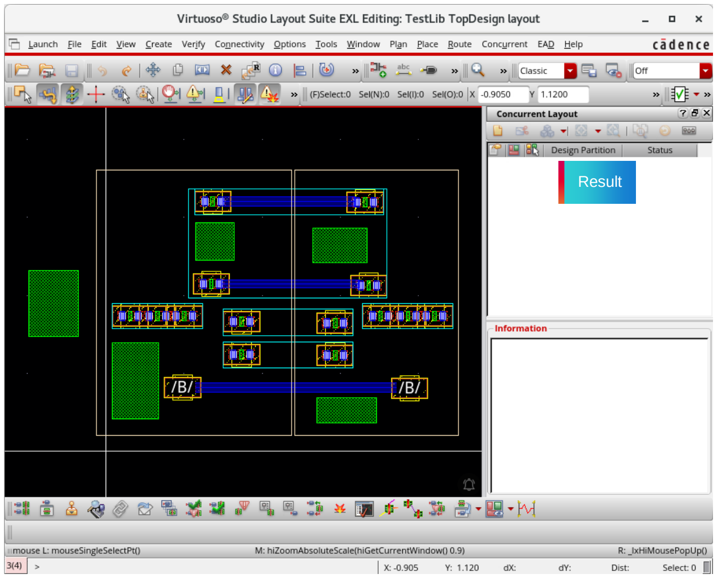
Result: Clearing the Design Partitions userA and userB in Manager Mode



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Module Summary

In this module, you learned how to

- List out the design partitions userA and userB in the top design
- Identify all the changes in the design partitions userA and userB in the top design
- Merge the submitted design partitions userA and userB
- Commit the merged design partitions userA and userB
- Clear the design partitions userA and userB in Manager mode

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Lab

Lab 6-1 Merging/Committing the Top Design in Manager Mode

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Quiz: Viewing the Changes of Peer Designers / Design Partitions

You can preview the changes from your peer designers and preview the design partitions from their top design to check their progress using the Preview button on the Concurrent Layout toolbar in the Concurrent Layout assistant.

A. True

B. False

Answer: A

Yes. You can preview the changes from your peer designers and preview the design partitions from their top design to check their progress using the Preview button on the Concurrent Layout toolbar in the Concurrent Layout assistant.

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Quiz: Reviewing All the Changes in the Design Partitions in the Top Design

You can review all the changes in detail in the design partitions in the top design using the Show Detailed Changes option in the Concurrent Layout assistant.

A. True

B. False

Answer: A

Yes. You can review all the changes in detail in the design partitions in the Top Design using the Show Detailed Changes option in the Concurrent Layout assistant.

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Quiz: Combining the Submitted Design Partitions

What options can you use to combine the submitted design partitions?

Select all that apply.

A. Merge
B. Merge/Un-Merge
C. Combine
D. Combine/Un-Combine
E. All of the above

Answer: A and B

You can use the Merge and Merge/Un-Merge options to merge the submitted design partitions.

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Quiz: Completing the Merge Process

You need to run the ______ command to complete the merge process and save the merged changes into the top design.

Enter the answer in the blank field.

Answer: Commit

You need to run the Commit command to complete the merge process and save the merged changes into the top design.

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Quiz: Deleting the Design Partitions

If your task is complete and there is no need for concurrent editing, you can delete the design partitions using the _____ All Design Partitions command.

Enter the answer in the blank field.

Answer: Clear

If your task is complete and there is no need for concurrent editing, you can delete the design partitions using the Clear All Design Partitions command.

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Module 7

Course Conclusions

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Summary

In this course, you learned to

- Set up and analyze the Concurrent Layout Editing environment
- Initialize and partition the top design in Manager mode
- Edit the design partitions in designer mode
- Merge and commit the top design in Manager mode

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References

Refer to this link for more details on the topics discussed in this course.

Virtuoso Concurrent Layout User Guide Product Version IC25.1

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Virtuoso Concurrent Layout User Guide Product Version IC23.1

Module 8

Next Steps

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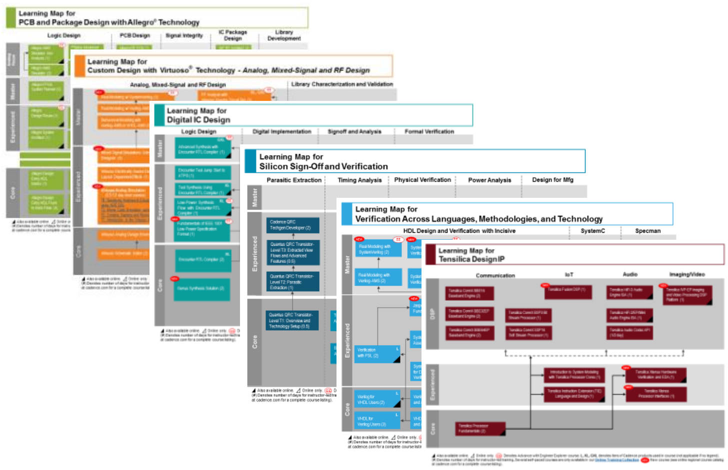

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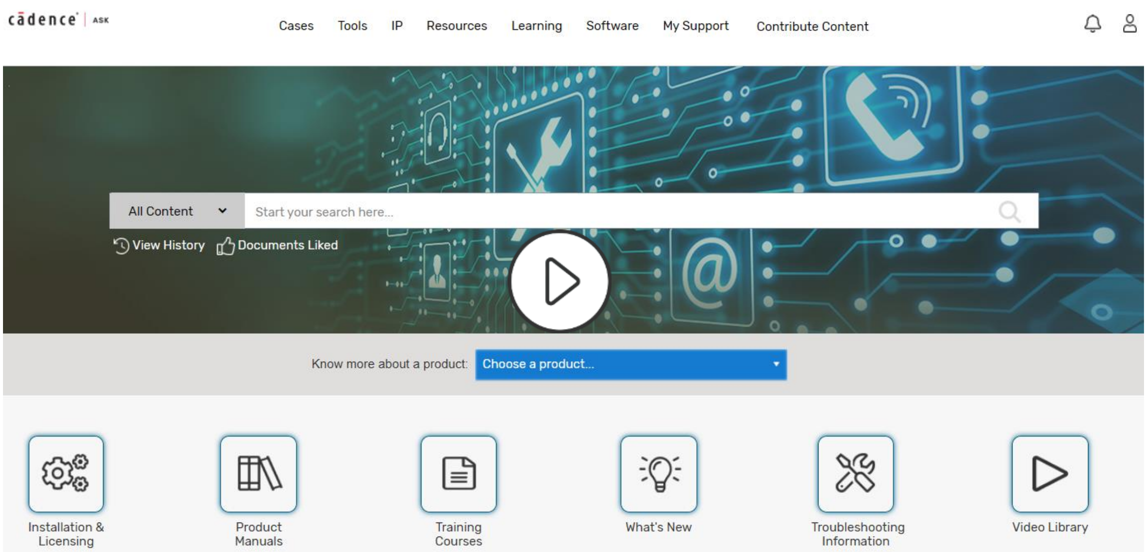

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Wrap Up

- Complete Post Assessment, if provided
- Complete the Course Evaluation
- Get a Certificate of Course Completion

Thank you!

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